The constant $A(p)$

Quasi-stationary amplitude, closed forms, and the Koenigs obstruction

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Contents

Chapter 2

The constant $A(p)$

2.1 Introduction

For the binary Galton–Watson process with division probability $p \in [0, \frac{1}{2})$ the survival probability decays geometrically, $S_n \sim A(p)(2p)^n$, and the mean population conditional on non-extinction converges to $1/A(p)$. That much was established in the mathematical introduction. Everything the limiting conditional law does — its mean, its variance, its dependence on how close the process sits to criticality — passes through the single constant

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n}, \qquad p \in [0, \tfrac{1}{2}). \tag{2.1}$$

This chapter is about that constant: what it is, how to compute it, why no elementary formula for it has been found despite a determined search, and exactly what can and cannot be proved about the failure of that search. The results are original to this thesis except where attributed.

Approximate values are easy enough to obtain. Iterate $S_{n+1} = 2pS_n - pS_n^2$ until S_n is very nearly stationary, then form $S_n/(2p)^n$. Repeating over a range of p traces the curve of figure 2.2, and this works well over most of the interval. It fails as $p \to \frac{1}{2}^-$, because the number of generations needed for S_n to reach near-stationarity grows without bound — from around 10^2 at $p = 0.2$ to nearly 5×10^5 at $p = 0.4999$.

After a short recapitulation of the setup inherited from the mathematical introduction (section 2.2), two exact representations come first. Rewriting the survival recursion as a telescoping product (section 2.3) proves that the limit (2.1) exists and is strictly positive; converting that product into a series (section 2.4) isolates the geometric decay in a factor carrying no dynamics at all, so that two-sided bounds can be read off by inspection. Those bounds trap $A(p)$ between half the continuous-time constant $A_c(p)$ and a companion upper bound, and a parity argument caps it at $\frac{1}{2}$. The same series then yields the near-critical behaviour $A(p) = 2\varepsilon[1 + O(\varepsilon \log(1/\varepsilon))]$, with $\varepsilon = 1 - 2p$ (section 2.5), locating the logarithm that makes convergence to the leading term slow, and with it the exact relationship between A and A_c (section 2.6).

What none of that yields is a formula. Section 2.7 records the search for one and the manner of its failure. The explanation is dynamical: under the substitution $S_n = 2w_n$ the survival recursion becomes the logistic map, and $A(p)$ is twice a value of that map's Koenigs linearising function (section 2.8); for every multiplier in the subcritical window that function is hypertranscendental (section 2.9), so no elementary expression in its own variable can be evaluated to produce $A(p)$. Section 2.10 delimits what this does and does not settle and reports the inverse-symbolic evidence bearing on what it leaves open, and sections 2.12 and 2.13 draw the practical and the residual conclusions.

2.2 Standing setup

This section fixes what is carried over from the mathematical introduction.

Let Z_n be the binary Galton–Watson process started from a single progenitor, $Z_0 = 1$, in which each individual independently divides into two with probability p and dies with probability $1-p$ at each generational increment. Its offspring generating function is $\phi(z) = pz^2 + (1-p)$, the mean population is $\mathbb{E}(Z_n) = (2p)^n$, and the survival probability $S_n = \mathbb{P}(Z_n > 0)$ satisfies

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1. \tag{2.2}$$

Throughout this chapter p is confined to the subcritical range

$$0 \le p < \tfrac{1}{2}, \tag{2.3}$$

where extinction is certain and $S_n \downarrow 0$. Two derived symbols are used constantly:

$$\varepsilon = 1-2p \in (0,1] \quad \text{(the distance from criticality)}, \qquad r = 2p \in [0,1) \quad \text{(the logistic multiplier)}. \tag{2.4}$$

Because the quadratic term in (2.2) becomes negligible once S_n is small, the decay is asymptotically geometric with ratio $2p$, and the constant $A(p)$ of (2.1) is the amplitude of that decay:

$$S_n \sim A(p)\,(2p)^n \qquad (n \to \infty). \tag{2.5}$$

The relation is an asymptotic equivalence, not an identity: S_n itself tends to zero, and what converges is the ratio $S_n/(2p)^n$. That this ratio converges to something finite and strictly positive is not assumed here; it is proved in section 2.3.

Splitting $\mathbb{E}(Z_n)$ over the events $\{Z_n = 0\}$ and $\{Z_n > 0\}$, the first contributing nothing, gives $\mathbb{E}(Z_n \mid Z_n > 0) = (2p)^n/S_n$ and hence

$$\mathbb{E}(Z_n \mid Z_n > 0) \;\longrightarrow\; \frac{1}{A(p)}, \tag{2.6}$$

the *quasi-stationary mean*; and the corresponding second-moment calculation gives a limiting conditional variance

$$\operatorname{Var}(Z_n \mid Z_n > 0) \;\longrightarrow\; \frac{1}{A(p)}\left(1 + \frac{1}{1-2p}\right) - \frac{1}{A(p)^2}, \tag{2.7}$$

again a function of p and $A(p)$ alone. The derivations of (2.6) and (2.7), together with the existence of the Yaglom limit they describe, are given in the mathematical introduction and are not repeated.

One comparison object is also inherited. The continuous-time linear birth–death process with per-capita birth rate λ and death rate $\mu > \lambda$ has a limiting conditional mean $\mu/(\mu-\lambda)$, and matching the parametrisations through the competing-clocks identity $p = \lambda/(\lambda+\mu)$ — proved in the mathematical introduction and used here by citation — turns this into

$$A_{\mathrm{c}}(p) = \frac{1-2p}{1-p} = \frac{2\varepsilon}{1+\varepsilon}, \qquad \frac{1}{A_{\mathrm{c}}(p)} = \frac{1-p}{1-2p}. \tag{2.8}$$

Unlike $A(p)$, the continuous-time constant $A_{\mathrm{c}}(p)$ is elementary.

Finally, an endpoint convention. At $p = 0$ the process dies at the first step and (2.2) degenerates; we set

$$A(0) := \lim_{p \downarrow 0} A(p) = \tfrac{1}{2}, \tag{2.9}$$

a value confirmed independently by the product of section 2.3 and by the parity bound of proposition 2.4.

2.3 An infinite-product representation

Take the nonlinear map for the survival probability,

$$S_{n+1} = 2pS_n - pS_n^2, \qquad S_0 = 1,$$

and substitute $S_n = 2w_n$. This gives

$$w_{n+1} = 2p\, w_n(1 - w_n) = r\, w_n(1 - w_n), \qquad r = 2p, \quad w_0 = \tfrac{1}{2}, \tag{2.10}$$

which is the logistic map — the same map that exhibits the period-doubling route to chaos as r increases from 0 to 4 (figure 2.1). The subcritical window is $r \in [0, 1)$, at the extreme left of that picture, where the origin is a globally attracting fixed point.

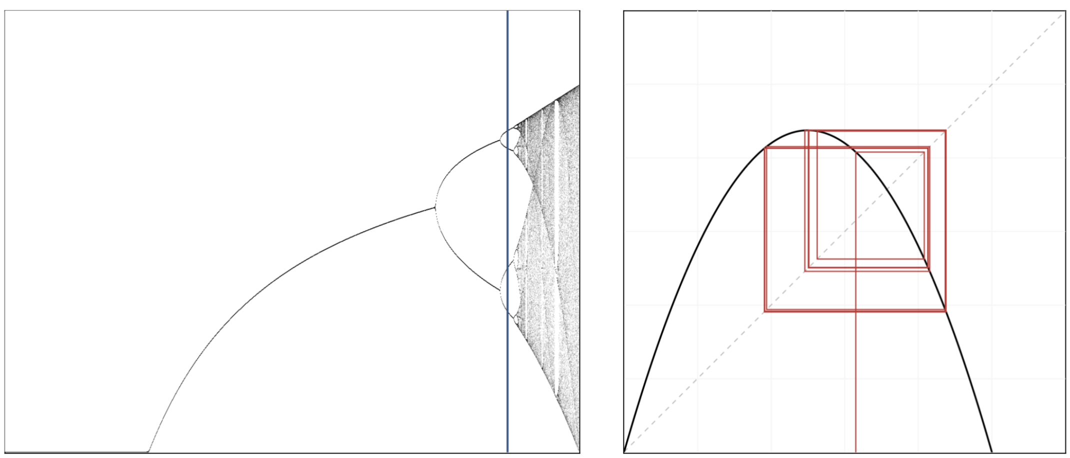


Figure 2.1: Bifurcation diagram of the logistic map $w \mapsto rw(1 - w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$; the right-hand plot shows a trajectory of the map at the same r.

In terms of w, (2.1) reads

$$A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} = 2 \lim_{n\to\infty} \frac{w_n}{r^n}. \tag{2.11}$$

Any sequence can be written as a telescoping product,

$$x_n = x_0 \left(\frac{x_1}{x_0}\right)\left(\frac{x_2}{x_1}\right) \cdots \left(\frac{x_n}{x_{n-1}}\right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k},$$

and applying this with $x_n = w_n / r^n$ the ratio simplifies at once:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{rw_k(1 - w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k. \tag{2.12}$$

The whole of the nonlinearity has been isolated into a single factor per generation, and the geometric decay has cancelled out entirely.

Proposition 2.1 (Product representation)**.** *For $p \in (0, \frac{1}{2})$, with S_n generated by (2.2) and $w_k = S_k/2$,*

$$A(p) = \prod_{k=0}^{\infty}(1 - w_k) = \frac{1}{2}\prod_{k=1}^{\infty}\left(1 - \frac{S_k}{2}\right), \tag{2.13}$$

and the product converges to a strictly positive limit. In particular the limit (2.1) defining $A(p)$ exists, and $0 < A(p) < \frac{1}{2}$.

Proof. Iterating (2.12) gives $x_n = x_0 \prod_{k=0}^{n-1}(1 - w_k)$ with $x_0 = w_0 = \frac{1}{2}$. For $0 < w_k < 1$ the infinite product $\prod_{k\geq 1}(1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k\geq 1} w_k < \infty$. From (2.10) with $0 < r < 1$ and $0 < w_n < 1$ we have $w_{n+1} < rw_n$, so by induction $w_n \leq w_0 r^n$; since $0 < r < 1$ this bound is summable, and the product converges to a positive finite value. Hence $\lim_n x_n$ exists and is positive, and by (2.11),

$$A(p) = 2\lim_n x_n = 2 \cdot \tfrac{1}{2}\prod_{k=0}^{\infty}(1 - w_k) = \prod_{k=0}^{\infty}(1 - w_k).$$

Pulling out the factor $1 - w_0 = \frac{1}{2}$ gives the second form, and since every remaining factor lies strictly in $(0, 1)$ for $p > 0$, the value is strictly below $\frac{1}{2}$. □

2.3.1 A certified truncation

The partial products

$$A_N(p) = \frac{1}{2}\prod_{k=1}^{N}\left(1 - \frac{S_k}{2}\right) \tag{2.14}$$

decrease monotonically to $A(p)$, so every truncation is a rigorous upper bound. A matching lower bound follows from the elementary inequality $\prod_k(1 - a_k) \geq 1 - \sum_k a_k$ for $a_k \in [0, 1]$ applied to the discarded tail. Since $w_{k+1} \leq rw_k$, the tail sum is dominated by a geometric series and

$$A_N(p)\left(1 - \frac{w_{N+1}}{1 - r}\right) \;\leq\; A(p) \;\leq\; A_N(p), \tag{2.15}$$

whenever the left-hand factor is positive.

2.3.2 Is this an advance?

Only a modest one. (2.13) is exact, it proves existence, and by (2.15) it brackets $A(p)$ at every order. But computing the factors still requires iterating the nonlinear map to generate the S_k, so the near-critical difficulty is untouched: as $r \to 1$ the bound $w_n \leq w_0 r^n$ degrades and the number of factors needed grows without limit.

2.4 An exact series and two-sided bounds

The product converts into a telescoping sum, and the sum is much more informative. Write $r = 2p$ and note that the recursion factorises as

$$S_{n+1} = rS_n\left(1 - \frac{S_n}{2}\right).$$

Define $v_n = r^n/S_n$, so that $v_n \to 1/A(p)$ by (2.1). Then

$$v_{n+1} - v_n = \frac{r^{n+1}}{rS_n\left(1-\frac{S_n}{2}\right)} - \frac{r^n}{S_n} = \frac{r^n}{S_n}\left[\frac{1}{1-\frac{S_n}{2}} - 1\right] = \frac{r^n}{S_n}\cdot\frac{S_n/2}{1-\frac{S_n}{2}} = \frac{r^n}{2-S_n}. \tag{2.16}$$

Summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 2.2 (Series representation)**.** *For $p \in [0, \frac{1}{2})$, with S_n generated by (2.2),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2-S_n}. \tag{2.17}$$

Unlike the product, this representation isolates the geometric decay in a factor that does not depend on the dynamics at all; the whole of the nonlinearity sits in the bounded denominator $2 - S_n$, which is confined to $[1, 2)$ since S_n lies in $(0, 1]$. Bounding the denominator by its extremes and summing the resulting geometric series gives the following.

Proposition 2.3 (Two-sided bounds)**.** *With $\varepsilon = 1 - 2p$ as in (2.4), for $p \in (0, \frac{1}{2})$,*

$$\frac{\varepsilon}{1+\varepsilon} \;<\; A(p) \;<\; \frac{2\varepsilon}{1+2\varepsilon}, \tag{2.18}$$

that is,

$$\frac{1-2p}{2(1-p)} \;<\; A(p) \;<\; \frac{2(1-2p)}{3-4p}. \tag{2.19}$$

The lower bound is exactly half the continuous-time constant (2.8):

$$A(p) > \tfrac{1}{2}A_{\mathrm{c}}(p). \tag{2.20}$$

Proof. Since $0 < S_n \le 1$ for all n we have $1 \le 2 - S_n < 2$, and hence $\frac{1}{2}r^n < r^n/(2-S_n) \le r^n$, with equality on the right only at $n = 0$, where $S_0 = 1$. Summing over $n \ge 0$ and using $\sum_n r^n = 1/(1-r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} \;<\; \frac{1}{A(p)} \;<\; 1 + \frac{1}{\varepsilon},$$

and inverting gives (2.18). Substituting $\varepsilon = 1 - 2p$ gives (2.19), whose lower bound is $(1-2p)/(2(1-p)) = \frac{1}{2}A_{\mathrm{c}}(p)$ by (2.8). ☐

At the far subcritical end, $p \to 0$, we have $\varepsilon \to 1$ and the lower bound tends to $\frac{1}{2}$. Since $A(p) \le \frac{1}{2}$ by proposition 2.4 below, the bound is asymptotically attained and $A(0^+) = \frac{1}{2}$, confirming the endpoint convention (2.9). At the critical end, $p \to \frac{1}{2}^-$, we have $\varepsilon \to 0$, the lower bound behaves like ε and the upper like 2ε; section 2.5 shows that it is the upper bound that is attained there.

Proposition 2.4 (Parity bound)**.** *For the binary Galton–Watson process started from a single individual, $A(p) \le \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \to 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so Z_n is even for every $n \ge 1$. Conditional on survival, therefore, $Z_n \ge 2$, and hence $\mathbb{E}(Z_n \mid Z_n > 0) \ge 2$ for all $n \ge 1$. Letting $n \to \infty$ and using (2.6) gives $1/A(p) \ge 2$. As $p \to 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit. ☐

The parity bound is a genuinely different kind of constraint from proposition 2.3: it comes from the arithmetic of the offspring law rather than from the analysis of the recursion, and it is what fixes the left-hand endpoint. Table 2.1 collects numerical values against all three bounds.

Table 2.1: The constant $A(p)$ computed from the product (2.13) in 60-digit arithmetic, against the bounds of proposition 2.3 and the near-critical asymptotic $2(1-2p)$ of theorem 2.5. The last two columns give the quasi-stationary mean and the number of factors of (2.13) needed for the product to reach full precision — the cost that section 2.12 has to manage.

p	$\frac{1}{2}A_c(p)$	$A(p)$	$\dfrac{2(1-2p)}{3-4p}$	$2(1-2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

2.5 Behaviour as $p \to \frac{1}{2}^-$

The bounds (2.18) show that $A(p)$ is of order ε near criticality but leave a factor of two unresolved. The series (2.17) settles it. It also identifies the order of the first relative correction, which turns out to carry a logarithm and is the reason the leading asymptotic is a poor surrogate except very close to $p = \frac{1}{2}$.

Theorem 2.5 (Near-critical amplitude)**.** *As $p \uparrow \frac{1}{2}$, with $\varepsilon = 1-2p \downarrow 0$,*

$$A(p) = 2\varepsilon \left[1 + O\left(\varepsilon \log \frac{1}{\varepsilon}\right)\right]. \tag{2.21}$$

In particular

$$A(p) \sim 2(1-2p), \qquad \frac{1}{A(p)} \sim \frac{1}{2(1-2p)}. \tag{2.22}$$

Proof. Write $r = 1-\varepsilon$. The identity

$$\frac{1}{2-S_n} = \frac{1}{2} + \frac{S_n}{2(2-S_n)} \tag{2.23}$$

turns the exact series (2.17) into

$$\frac{1}{A(p)} = 1 + \frac{1}{2\varepsilon} + D_\varepsilon, \qquad D_\varepsilon := \sum_{n=0}^{\infty} r^n \frac{S_n}{2(2-S_n)}, \tag{2.24}$$

the geometric sum $\sum_n r^n = 1/\varepsilon$ having been evaluated. Only a logarithmic bound on the remainder D_ε is needed.

The map $s \mapsto ps(2-s)$ is increasing in p and in s on $s \in [0,1]$, so comparing (2.2) at parameter p with the same recursion at $p = \frac{1}{2}$ from the common initial value $S_0 = 1$ gives, by induction, $S_n(p) \le S_n(\frac{1}{2})$ for every n. At criticality the reciprocal increments satisfy

$$\frac{1}{S_{n+1}(\frac{1}{2})} - \frac{1}{S_n(\frac{1}{2})} = \frac{1}{2(1-S_n(\frac{1}{2})/2)} \ \ge\ \frac{1}{2}, \tag{2.25}$$

so $1/S_n(\frac{1}{2}) \geq 1 + n/2$ and hence $S_n(\frac{1}{2}) \leq 2/(n+2)$. Since $2 - S_n \geq 1$, these two facts combine to give

$$\begin{aligned} 0 \leq D_\varepsilon \leq \frac{1}{2}\sum_{n=0}^{\infty} r^n S_n(p) \ &\leq \ \frac{1}{2}\sum_{n=0}^{\infty} r^n \, \frac{2}{n+2} \ = \ \sum_{n=0}^{\infty} \frac{r^n}{n+2} \\ &= \frac{-\log(1-r) - r}{r^2} \ = \ O\left(\log \frac{1}{\varepsilon}\right). \end{aligned} \tag{2.26}$$

Factoring $1/(2\varepsilon)$ out of (2.24),

$$\frac{1}{A(p)} = \frac{1}{2\varepsilon}\Big[1 + 2\varepsilon(1 + D_\varepsilon)\Big],$$

and the quantity following the leading 1 is $O(\varepsilon \log(1/\varepsilon))$. Inverting gives (2.21). □

The proof locates the logarithm precisely: it comes from the near-critical decay $S_n \lesssim 2/(n+2)$ persisting across the whole range $n \lesssim 1/\varepsilon$ over which the geometric factor r^n still carries weight. The summand $r^n/(2 - S_n)$ involves two decays on very different scales: the geometric factor $r^n = (1-\varepsilon)^n$ decays on the scale $n \sim 1/\varepsilon$, which is large, whereas S_n is close to its critical behaviour $2/n$ and has already fallen to near zero after $O(1)$ generations. For the overwhelming majority of the terms that carry weight, therefore, $S_n \approx 0$ and the summand is approximately $r^n/2$; summing gives $1/A \approx 1/(2\varepsilon)$ at once. The rigorous argument above is that estimate with the neglected mass bounded rather than discarded.

Remark 2.6 (The critical gradient). By (2.22), $A(p)$ meets the axis at $p = \frac{1}{2}$ with gradient -4:

$$A(p) = 2 - 4p + o(1 - 2p) \qquad (p \to \tfrac{1}{2}^-). \tag{2.27}$$

This is the single most useful diagnostic in section 2.7: since $1/A$ diverges as $A \to 0$, an error in A of vanishing absolute size still produces an unbounded error in the quasi-stationary mean, so a candidate accurate to plotting resolution across the whole interval can fail here, and fail fatally.

Remark 2.7 (Rate of approach). Numerically, $A(p)/2\varepsilon$ approaches 1 rather slowly. Evaluating (2.13) in high precision gives $A/2\varepsilon = 0.90987$, 0.98612, 0.99814, 0.99977 and 0.999972 at $\varepsilon = 2 \times 10^{-2}$ down to 2×10^{-6}. An exploratory fit to the deficit gives

$$1 - \frac{A}{2\varepsilon} \approx \varepsilon\left(\log\frac{1}{\varepsilon} + c\right), \qquad c \approx 0.78, \tag{2.28}$$

consistent with (2.21). The constant c is empirical.

Why the continuum route is worse

The same leading asymptotic can be reached through the Riccati approximation of the mathematical introduction. The exact difference is

$$S_{n+1} - S_n = (2p-1)S_n - pS_n^2 = -\varepsilon S_n - \tfrac{1}{2}S_n^2 + \tfrac{\varepsilon}{2}S_n^2,$$

and dropping the $O(\varepsilon S_n^2)$ term gives $\mathrm{d}S/\mathrm{d}n = -\varepsilon S - \frac{1}{2}S^2$. Setting $u = 1/S$ linearises this to $u' = \varepsilon u + \frac{1}{2}$, with solution

$$u(n) = C\mathrm{e}^{\varepsilon n} - \frac{1}{2\varepsilon}, \qquad S(n) = \frac{1}{C\mathrm{e}^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim A\mathrm{e}^{-\varepsilon n}$ for small ε identifies $A = 1/C$, and matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$, recovering (2.22). Both steps are heuristic and the second is delicate; worse, the route gives no control at all on the error, and so cannot see the logarithm of (2.21).

2.6 Why the discrete and continuous constants differ

The two constants are

$$A_{\mathrm{c}}(p) = \frac{1-2p}{1-p} \quad \text{(continuous time)}, \qquad A(p) = \lim_{n\to\infty} \frac{S_n}{(2p)^n} \quad \text{(discrete time)},$$

and the observation recorded in the mathematical introduction was that the first starts at 1 where the second starts at $\frac{1}{2}$, yet halving the first does not superpose the curves (figure 2.2).

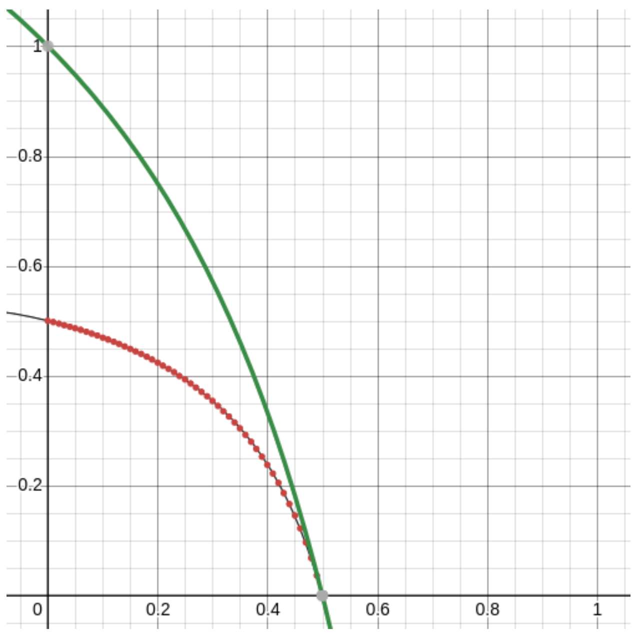


Figure 2.2: The continuous-time constant $A_{\mathrm{c}}(p) = (1-2p)/(1-p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by a constant factor in between. The black line is a formula obtained by symbolic regression (section 2.7); it tracks the points closely but is not exact.

What holds is not $A = \frac{1}{2}A_{\mathrm{c}}$ but the strict inequality $A > \frac{1}{2}A_{\mathrm{c}}$ of (2.20), and the inequality is attained only in the limit $p \to 0$. Combining (2.8) and (2.22), the ratio runs monotonically across the interval:

$$\frac{A(p)}{A_{\mathrm{c}}(p)} \longrightarrow \frac{1}{2} \quad (p \to 0), \qquad \frac{A(p)}{A_{\mathrm{c}}(p)} \longrightarrow 1 \quad (p \to \tfrac{1}{2}^-), \tag{2.29}$$

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$.

At $p \to 0$ the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so a surviving population is always even (proposition 2.4); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model a survivor is a single individual that has not yet died, so the quasi-stationary mean is 1. The reciprocals are $\frac{1}{2}$ and 1.

At $p \to \frac{1}{2}^{-}$ the factor disappears, because both constants are governed by the same critical scaling. Expanding (2.8) near $p = \frac{1}{2}$ gives $A_{\mathrm{c}}(p) = 2\varepsilon/(1+\varepsilon) \to 2\varepsilon$, which is precisely (2.22). Near criticality the discreteness of the generational clock stops mattering; what survives is the universal 2ε behaviour, and the two models agree to leading order. They do not agree at the next order: the continuous constant has the elementary expansion $A_{\mathrm{c}} = 2\varepsilon(1 - \varepsilon + O(\varepsilon^2))$, whereas theorem 2.5 shows the discrete correction carries a factor $\log(1/\varepsilon)$ and is therefore asymptotically larger.

Where the closed form went

In continuous time the survival probability is elementary at every time t, not merely asymptotically:

$$S(t) = \frac{\mu - \lambda}{\mu - \lambda \mathrm{e}^{-(\mu-\lambda)t}} \cdot \frac{\mu}{\mu} \quad \text{rearranges to} \quad S(t) = \frac{\mu - \lambda}{\mu} \mathrm{e}^{-(\mu-\lambda)t} \left(1 - \frac{\lambda}{\mu} \mathrm{e}^{-(\mu-\lambda)t}\right)^{-1}, \tag{2.30}$$

so the amplitude $(\mu - \lambda)/\mu$ can simply be read off the exact solution. In discrete time no such exact solution exists: the survival probability is defined by iterating a quadratic map, and its amplitude is the infinite product (2.13) of every early correction. The continuous process is solvable because its generating function satisfies a linear first-order partial differential equation with an elementary characteristic; the discrete one is not because iterating a quadratic map is, in the precise sense of section 2.9, a transcendental operation.

2.7 Searching for a closed form

Several methods were tried in the hope of an elementary formula for $A(p)$. The first was the continuum route of section 2.5: treat the map as a differential equation in the large-generation limit and solve it. This gives the shape of the decay but leaves the constant of integration undetermined, which is to say it leaves $A(p)$ undetermined.

The second was symbolic regression by genetic algorithm. The idea is simple: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those; then measure again, and repeat the cycle of variation and selection many times over [1, 2]. The first implementation searched a space of polynomials, on the hunch that the answer might take that form and because such functions are simple to encode: a genome is a coefficient list $[a_0, \ldots, a_k]$ for $a_0 + a_1 p + \cdots + a_k p^k$ up to some maximum order, drawn uniformly at random. It became apparent quickly that many substantially different formulae produce curves indistinguishable from the target by eye. Supposing that a true formula might have rational coefficients, a second version searched $\sum_i (a_i/b_i)p^i$ with $[a_i], [b_i] \in \mathbb{Z}^{k+1}$, and a third extended the space to ratios of two such polynomials. Candidates could be constrained to the known endpoints $A(0) = \frac{1}{2}$ and $A(\frac{1}{2}) = 0$, and the best that emerged fitted the high-precision numerical data well:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \tag{2.31}$$

This passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$ and looks convincing, but it fails the diagnostic of remark 2.6: differentiating at $p = \frac{1}{2}$ gives

$$\hat{A}_1'(\tfrac{1}{2}) = -\frac{2}{0.55} \approx -3.636$$

rather than the required -4.

A later attempt used a symbolic tree search over progressively nested elementary functions of all kinds. The conclusion was the same. Many functions of widely varying form resemble the target closely, and no satisfactory simple formula emerges even when candidates with fewer components are explicitly favoured. The expressions returned were typically deep nestings of trigonometric and iterated exponential terms carrying a dozen or more fitted decimal constants, and they were not interpretable in any useful sense; a representative and comparatively concise one is

$$\hat{A}_3(p) = \frac{-0.616647881739505\, \mathrm{e}^{-0.552605335919693\, \mathrm{e}^{p}}}{\sin(\cos p) - |p|} + 0.921964323679771. \tag{2.32}$$

These are records of a curve-fitting exercise, not proposed identities.

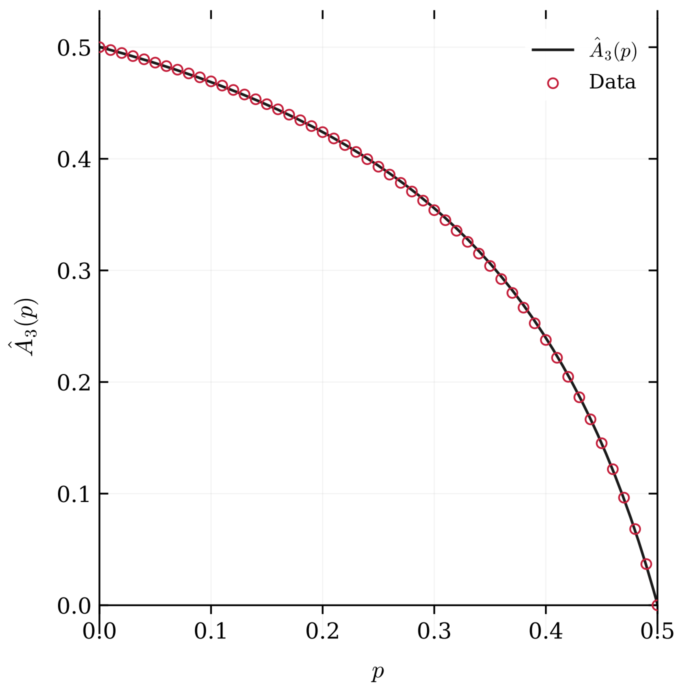


Figure 2.3: The symbolic-regression candidate $\hat{A}_3(p)$ of (2.32) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point, returning $\hat{A}_3(\frac{1}{2}) = 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient -3.63 instead of -4. Errors of this size are invisible here but are fatal for $1/A$ near criticality.

The recurring defect is the same in every case. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4, and by remark 2.6 that is fatal near criticality.

A pragmatic patch is to splice the asymptotic onto a fitted curve,

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545\, \mathrm{e}^{\mathrm{e}^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962, \end{cases} \tag{2.33}$$

and this does work as a surrogate. But if one is going to splice, it is better to splice onto something exact: the certified partial product (2.15) on the first branch and the proved asymptotic (2.21) on the second. Section 2.12 does exactly that.

That an enormous number of very different formulae can behave almost identically over a bounded interval is not merely an inconvenience of the method; it is a symptom. Curve fitting cannot distinguish a function that has a closed form from one that does not, and no quantity of fitting will ever settle the question.

2.8 The Koenigs identification

A dynamical argument explains why no elementary closed form should be expected. In short: $A(p)$ is a value of the Koenigs linearising function of the logistic map, and that function is known to be non-elementary throughout the parameter range required.

2.8.1 From the survival recursion to the logistic map

The substitution has already been made in section 2.3. With $S_n = 2w_n$ and $r = 2p$, the survival recursion (2.2) becomes the logistic iteration

$$w_{n+1} = f_r(w_n), \qquad f_r(w) = r\,w(1-w), \qquad w_0 = \tfrac{1}{2}, \tag{2.34}$$

with $r \in [0, 1)$ throughout the subcritical regime, and

$$A(p) = 2 \lim_{n\to\infty} \frac{w_n}{r^n}. \tag{2.35}$$

2.8.2 Koenigs linearisation

Definition 2.8 (Koenigs function)**.** Let $f_r(z) = rz(1-z)$ have a fixed point at $z = 0$ with multiplier $r = f_r'(0)$ satisfying $0 < |r| < 1$. Koenigs [3] proved that there is a unique function ψ_r, holomorphic in a neighbourhood of the origin, satisfying the Schröder equation

$$\psi_r\big(f_r(z)\big) = r\,\psi_r(z), \qquad \psi_r(0) = 0, \qquad \psi_r'(0) = 1. \tag{2.36}$$

Equivalently,

$$\psi_r(z) = \lim_{n\to\infty} r^{-n} f_r^{\circ n}(z) \tag{2.37}$$

uniformly near the origin. The normalisation $\psi_r'(0) = 1$ is equivalent to $\psi_r(z)/z \to 1$ as $z \to 0$.

The Koenigs function is a change of coordinate that straightens the dynamics: in the ψ coordinate the nonlinear map f_r becomes multiplication by r (figure 2.4). Here f_r is exactly (2.34), with $z = w_n$ and $f_r(z) = w_{n+1}$, and the multiplier is $r = 2p \in (0, 1)$ — an attracting fixed point, which is the classical Koenigs setting. Standard accounts are Milnor [4] and Carleson and Gamelin [5].

2.8.3 The identity

Koenigs' theorem is local: the series (2.37) is guaranteed to converge only on a disc about the origin, and for the logistic map that disc has radius $(1-r)/r$. The evaluation point of interest is $w_0 = \frac{1}{2}$, and $\frac{1}{2} < (1-r)/r$ fails as soon as $r \geq \frac{2}{3}$, that is $p \geq \frac{1}{3}$. For those parameters $\frac{1}{2}$ lies outside the germ, and writing "$\psi_r(\frac{1}{2})$" as a naive evaluation of a power series would be meaningless.

The remedy is the functional equation itself. For $r \in (0, 1)$ the origin is attracting on the real line and the whole interval $(0, 1)$ lies in its real basin, so the orbit $w_n = f_r^{\circ n}(w_0)$ enters the germ after finitely many steps. Define the *basin extension*

$$\Psi_r(w) := \lim_{n\to\infty} r^{-n} f_r^{\circ n}(w), \qquad w \in (0, 1), \tag{2.38}$$

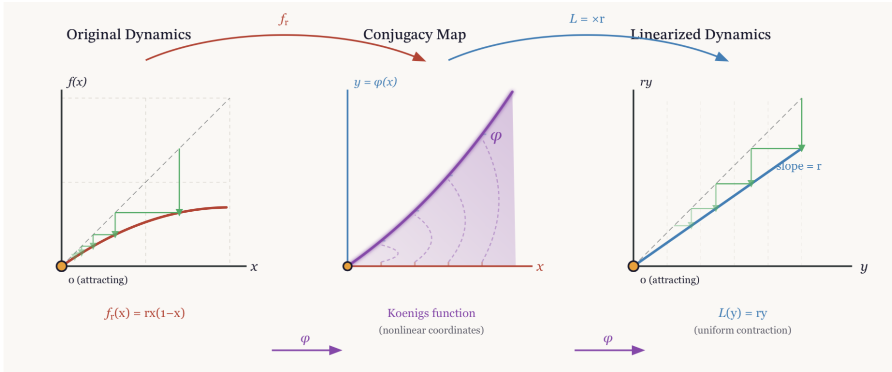


Figure 2.4: Koenigs linearisation of the logistic map near an attracting fixed point. The nonlinear dynamics $x \mapsto f_r(x) = rx(1-x)$ are conjugated by ψ_r to the linear map $y \mapsto ry$; the diagram displays the functional relation $\psi_r(f_r(x)) = r\psi_r(x)$ and the role of ψ_r as the coordinate change that straightens the dynamics.

equivalently $\Psi_r(w) = r^{-N}\psi_r(f_r^{\circ N}(w))$ for any N large enough that $f_r^{\circ N}(w)$ lies in the germ. The two definitions agree where both apply, because (2.36) makes the right-hand side independent of N, and Ψ_r is the unique analytic continuation of ψ_r along the orbit.

Proposition 2.9 (The constant as a Koenigs value)**.** *For $p \in (0, \frac{1}{2})$ and $r = 2p$,*

$$A(p) = 2\,\Psi_r\left(\tfrac{1}{2}\right) = 2\lim_{n\to\infty} r^{-n} f_r^{\circ n}\left(\tfrac{1}{2}\right). \tag{2.39}$$

Proof. Iterating (2.36) along the orbit $w_0, w_1, w_2, \ldots$ of (2.34), from any point at which the germ applies, gives $\psi_r(w_n) = r\,\psi_r(w_{n-1})$ and hence

$$\Psi_r(w_0) = \frac{\psi_r(w_n)}{r^n} \qquad \text{for every } n \text{ large enough that } w_n \text{ lies in the germ.} \tag{2.40}$$

Passing to the limit, $\Psi_r(w_0) = \lim_n \psi_r(w_n)/r^n$. The normalisation $\psi_r(0) = 0$, $\psi_r'(0) = 1$ means $\psi_r(z)/z \to 1$ as $z \to 0$, and $w_n = S_n/2 \to 0$ because $r < 1$; replacing $\psi_r(w_n)$ by w_n in the limit is therefore legitimate, and $\Psi_r(w_0) = \lim_n w_n/r^n$. Comparing with (2.35) and using $w_0 = S_0/2 = \frac{1}{2}$ gives $A(p) = 2\Psi_r(\frac{1}{2})$. □

We write $A(p) = 2\psi_r(\frac{1}{2})$ below as shorthand for this basin value.

Remark 2.10 (Germ versus basin). Differential algebraicity is a local condition preserved under analytic continuation: if an algebraic differential equation held for ψ_r on any subdomain, it would persist along the continuation defining Ψ_r, and conversely. Hypertranscendence of the germ and of the basin extension are therefore equivalent statements, and the identity (2.39) is unaffected by which one is under discussion.

2.9 No elementary conjugacy in the subcritical window

This section makes the obstruction precise. The logical skeleton is short:

(i) every elementary function satisfies an algebraic differential equation;

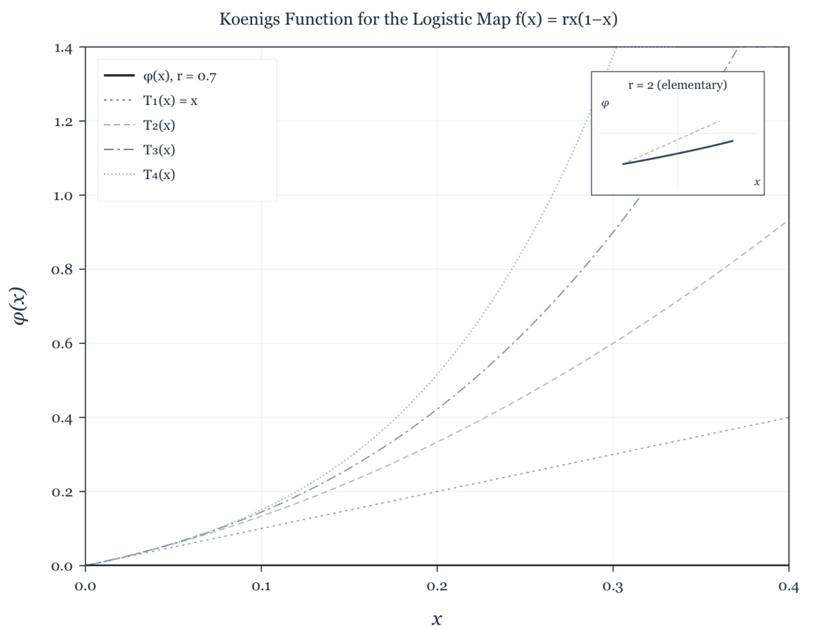


Figure 2.5: Numerical approximation of a Koenigs function for a logistic multiplier in $(0, 1)$, with low-order Taylor polynomials near the origin. Elementary closed forms exist only for exceptional multipliers outside this range; those cases are derived in section 2.A.

(ii) differential algebraicity survives compositional inversion;

(iii) by a theorem of Becker and Bergweiler, a solution of Schröder's equation for a rational map of degree at least two can be differentially algebraic only when the underlying fixed point is *repelling*;

(iv) for $r \in (0, 1)$ the fixed point 0 of f_r is attracting.

Hypertranscendence of ψ_r at every subcritical parameter follows at once — with no exceptional cases to exclude, because the attracting window contains no candidates at all. Section 2.10 then explains what this does *not* settle.

2.9.1 Differential algebraicity, hypertranscendence, elementarity

Definition 2.11 (Differentially algebraic; hypertranscendental)**.** A function g, holomorphic on a domain, or a formal power series, is *differentially algebraic over* $\mathbb{C}(z)$ if there is a non-zero polynomial P in $n + 2$ variables, for some $n \geq 0$, with

$$P\big(z,\, g(z),\, g'(z),\, \ldots,\, g^{(n)}(z)\big) \equiv 0. \tag{2.41}$$

Otherwise g is *differentially transcendental*, or *hypertranscendental*.

Lemma 2.12 (Elementary implies differentially algebraic)**.** *Every elementary function in the sense of Liouville — any function reachable from $\mathbb{C}(z)$ by a finite tower of algebraic operations, exponentials and logarithms — is differentially algebraic. Consequently a hypertranscendental function is not elementary.*

Proof sketch. Each storey of a Liouvillian tower preserves differential algebraicity: algebraic extensions do so by elimination; if g is differentially algebraic then so are e^g and $\log g$, their defining first-order relations combining with the equation for g; and the class is closed under the field operations and under composition. These closure properties are classical; see Rubel's survey [6]. $\square$

Lemma 2.13 (Closure under compositional inversion). *Let g be holomorphic near 0 with $g(0) = 0$ and $g'(0) \neq 0$, and let g^{-1} denote its local compositional inverse. Then g is differentially algebraic over $\mathbb{C}(z)$ if and only if g^{-1} is (Moore [7]; see also Boshernitzan and Rubel [8]).*

Lemma 2.13 matters because the literature states Schröder-equation results in two equivalent but opposite conventions, and conflating them is an easy way to misquote a theorem. Our ψ_r is the *linearising coordinate*: $\psi_r(f_r(z)) = r\psi_r(z)$. The function classified by Becker and Bergweiler — and by Ritt before them — is the *linearising parametrisation* $\varphi_r = \psi_r^{-1}$, which satisfies

$$\varphi_r(r\,t) = f_r(\varphi_r(t)), \qquad \varphi_r(0) = 0, \quad \varphi_r'(0) = 1. \tag{2.42}$$

By lemma 2.13, hypertranscendence of either function transfers to the other.

2.9.2 The Becker–Bergweiler theorem, stated precisely

The result needed is the 1995 classification of differentially algebraic Schröder functions [9]; we quote it in the form given in Fernandes' survey of the Schröder, Böttcher and Abel equations [10, Thm 3.1].

Theorem 2.14 (Becker–Bergweiler [9], as restated in [10, Thm 3.1]). *Let R be a rational map of degree at least two with $R(0) = 0$ and multiplier $s = R'(0)$, where $s \neq 0$ and s is not a root of unity; for $|s| = 1$ the Schröder function is understood as a formal series. Let φ, normalised by $\varphi(0) = 0$ and $\varphi'(0) = 1$, solve $\varphi(st) = R(\varphi(t))$. If φ is differentially algebraic over $\mathbb{C}(t)$, then*

(i) *the fixed point 0 is* repelling, *$|s| > 1$; and*

(ii) *φ is a Möbius transformation of one of*

$$\exp(\alpha t^{\varrho}), \qquad \cos(\alpha t^{\varrho} + \beta), \qquad \wp(\alpha t^{\varrho} + \beta),\ \wp^2,\ \wp^3,\ \wp'(\alpha t^{\varrho} + \beta),$$

where $\wp$ is a Weierstrass function, $\varrho \in \mathbb{Q}$ is such that the function concerned is meromorphic on $\mathbb{C}$, $\alpha \neq 0$, and β is a fraction of a period. Correspondingly R is Möbius-conjugate to $z^{\pm d}$, to $\pm T_d$ (Chebyshev), or to a Lattès map.

Remark 2.15 (Lineage and independent corroboration). For $|s| > 1$ — the Poincaré-function case — the classification goes back to Ritt in 1926 [11]; Becker and Bergweiler completed the picture for the three classical functional equations at once, and Bergweiler's later account with Aschenbrenner [12] assigns the exceptional list explicitly to the repelling case. Di Vizio and Fernandes [13] have given an independent difference-Galois proof of Ritt's theorem; notably, their Theorems 1.2 and 4.2 assume only that s is neither zero nor a root of unity — no condition $|s| > 1$ — and show that any differentially algebraic solution of $y(R(x)) = s\,y(x)$, which is our ψ-convention with no inversion needed, must already satisfy a first-order equation of Ritt type $(y')^{\varrho} = B(x)y^{j}$.

2.9.3 Hypertranscendence throughout the subcritical window

Theorem 2.16 (No exceptional parameters in $(0,1)$)**.** *For every $r \in (0,1)$, the Koenigs function ψ_r of $f_r(z) = rz(1-z)$ at 0 is hypertranscendental over $\mathbb{C}(z)$: it satisfies no algebraic differential equation. In particular ψ_r is not an elementary function.*

Proof. Fix $r \in (0,1)$. The map f_r is rational of degree two, fixes 0, and its multiplier $s = f_r'(0) = r$ satisfies $0 < s < 1$; the hypotheses of theorem 2.14 hold, with 0 attracting. Suppose for contradiction that ψ_r were differentially algebraic. Its local inverse $\varphi_r = \psi_r^{-1}$ solves the parametrisation form (2.42) with the stated normalisation, and by lemma 2.13 it would be differentially algebraic too. Theorem 2.14(i) would then force $|r| > 1$, contradicting $r \in (0,1)$. Hence ψ_r is hypertranscendental, and non-elementarity follows from lemma 2.12. □

Remark 2.17 (The exceptional set inside the window is empty, not merely avoided)*.* The affine conjugacy $c(r) = (2r - r^2)/4$ of lemma 2.18 sends $r \in (0,1)$ to $c \in (0, \frac{1}{4})$, which avoids the exceptional quadratic parameters $c = 0$ (reached at $r = 2$) and $c = -2$ (reached at $r = 4$, and also at $r = -2$). The exceptional Schröder functions exist only over *repelling* fixed points, so the attracting window contains no candidates whatsoever, and no enumeration of exceptional parameters is required. Consistently, the classical elementary cases of section 2.A have multipliers 2 and 4 at the linearised fixed point — both repelling: they are Poincaré-type linearisations, and neither can be pressed into service for $r \in (0,1)$.

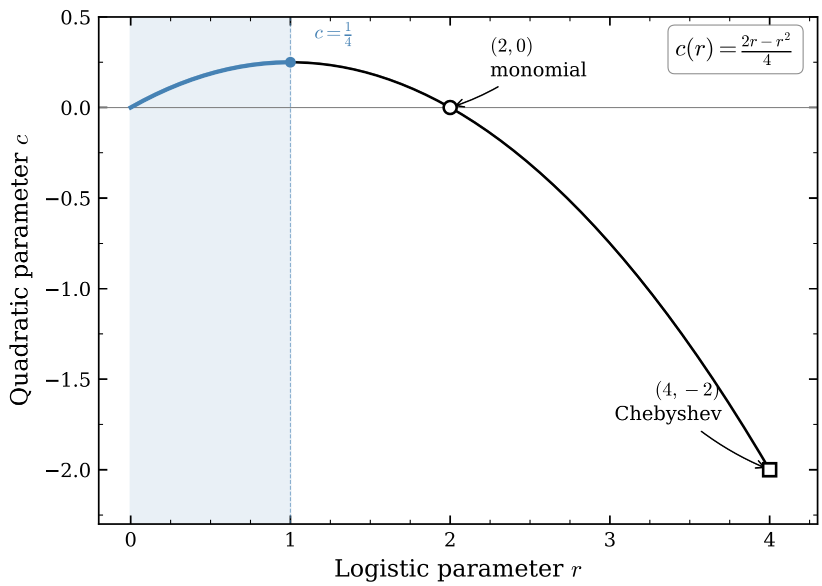


Figure 2.6: The affine parameter correspondence $c(r) = (2r - r^2)/4$ between the logistic family $f_r(x) = rx(1-x)$ and the quadratic family $P_c(z) = z^2 + c$. The shaded segment is $r \in (0,1)$, the subcritical branching window, which maps to $c \in (0, \frac{1}{4})$; the two integrable parameters, $r = 2$ ($c = 0$, monomial) and $r = 4$ ($c = -2$, Chebyshev), lie outside it.

2.9.4 Where the window sits in the quadratic family

The main cardioid of the Mandelbrot set — the set of c for which $z^2 + c$ has an attracting fixed point — is parametrised by that point's multiplier λ through $c = \lambda/2 - \lambda^2/4$, which is exactly the correspondence of lemma 2.18 with $\lambda = r$. The branching window $r \in (0, 1)$ is therefore the real diameter of the main cardioid, traversed from the centre $c = 0$ out to the parabolic cusp $c = \frac{1}{4}$, where $\lambda = 1$ and the process turns critical. Figure 2.7 places the segment in context.

Throughout the window the Julia set is a quasicircle bounding the basin of the attracting fixed point, and it is a genuine fractal for every c in the interval. The Koenigs function is the conformal coordinate that linearises the dynamics inside that basin, so its natural domain is bounded by exactly this curve. A function whose domain of definition has no smooth description is not a promising candidate for a closed form, and $A(p)$, being one of its values, inherits the difficulty. The contrast with the integrable cases is sharp: at $c = 0$ the Julia set degenerates to the unit circle, and it is precisely there that the linearisation becomes elementary, (2.45). The picture illustrates theorem 2.16; it does not prove it.

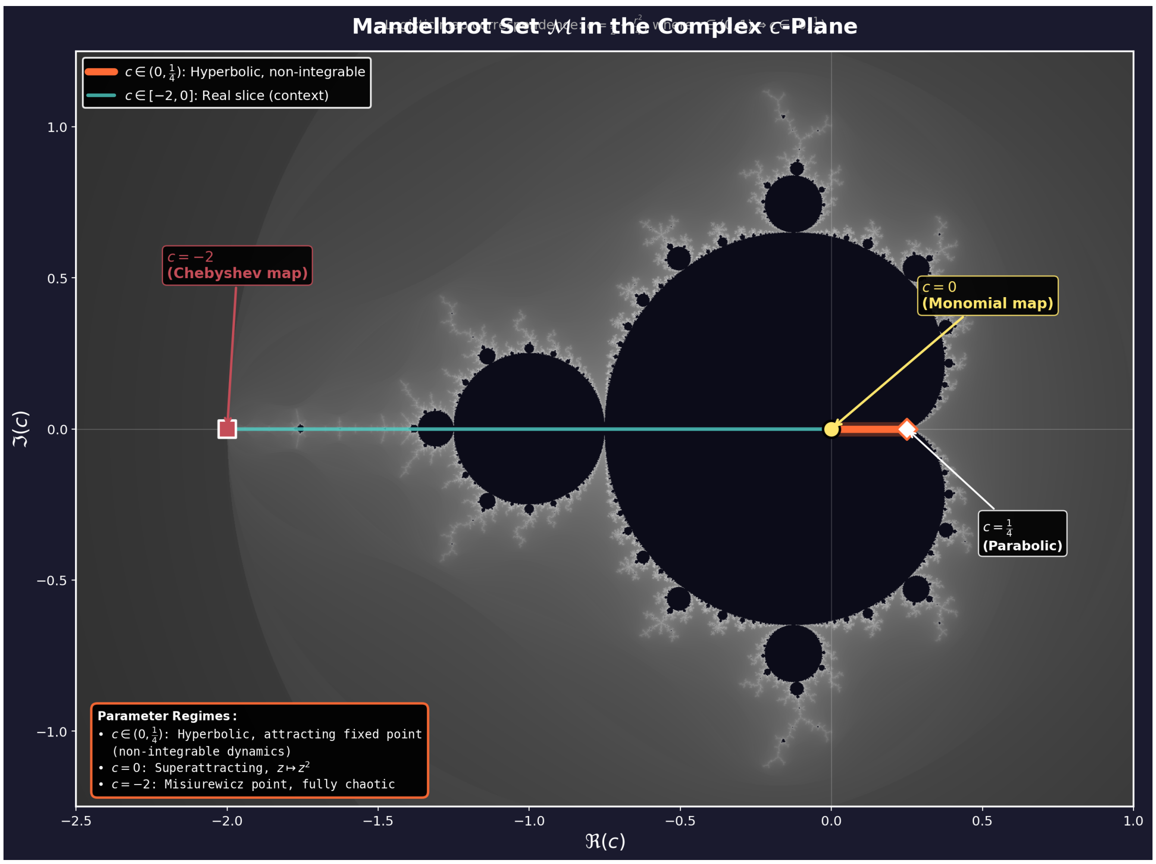


Figure 2.7: The Mandelbrot set in the complex c-plane, with the interval $c \in (0, \frac{1}{4})$ highlighted. Under the correspondence $c = (2r - r^2)/4$ of lemma 2.18 this is the image of the subcritical branching window $r = 2p \in (0, 1)$, and it forms the real diameter of the main cardioid, running from the centre $c = 0$ to the parabolic cusp $c = \frac{1}{4}$, which is criticality. The two integrable parameters are marked: $c = 0$ (monomial, reached in logistic coordinates at $r = 2$) and $c = -2$ (Chebyshev, $r = 4$), the latter at the far tip of the real slice. Note that $c(r) = c(2 - r)$, so each $c \in (0, \frac{1}{4})$ is the image of two logistic parameters, r and $2 - r$; these correspond to the two fixed points of $z^2 + c$, with multipliers r and $2 - r$, and it is the attracting one, $r < 1$, that governs $A(p)$.

2.10 What is and is not proved about $A(p)$

"Closed form" must be made precise before any claim is stated, because at least four inequivalent readings are in play and a result for one is not a result for another:

(V) the *value* $A(p_0)$ is elementary, or algebraic, for a fixed rational p_0;

(E) the *map* $p \mapsto A(p)$ is an elementary function of p;

(D) the map is *differentially algebraic in* p, that is, it satisfies some algebraic ordinary differential equation in p;

(S) the map is expressible through standard special functions (Γ, ζ, ${}_2F_1$, theta or q-series).

Theorem 2.16 speaks to the function $z \mapsto \psi_r(z)$ at fixed r, and licenses exactly this much: *no evaluation of $A(p)$ can be short-circuited through an algebraic differential equation for the lineariser, at any subcritical parameter.* It does not, by itself, decide any of (V), (E), (D) or (S).

The value gap

A hypertranscendental function can take elementary — even algebraic — values at individual points: Γ is hypertranscendental by Hölder's theorem, and yet $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Nothing established so far forbids, say, $A(\frac{1}{4})$ from being an algebraic number.

There *is* a genuine values theorem in this circle of ideas. Becker and Bergweiler proved in 1993 [14] that the Böttcher conjugacy between two polynomials of the *same degree* $d \geq 2$ with algebraic coefficients, not linearly conjugate to monomials or Chebyshev polynomials, is transcendental and takes transcendental values at algebraic points; Mahler's method drives the proof (for the method see Nishioka [15]). One might hope to transfer this to the Koenigs value $\psi_r(\frac{1}{2})$. It does not transfer, for two independent reasons.

Formally, the Schröder equation pairs an inner map of degree two with an outer map — multiplication by r — of degree one, so the equal-degree hypothesis fails outright.

Structurally, no easy repair exists. Mahler's method balances doubly-exponential analytic decay along an orbit against comparable height growth, and in the Böttcher setting both exponents run like d^n. In the attracting Koenigs setting the orbit $w_n = f_r^{\circ n}(\frac{1}{2})$ decays only geometrically, $|w_n| \sim A\, r^n$, while the heights of the rational numbers w_n still grow doubly exponentially, $h(w_n) \sim C\, 2^n$; the resulting inequalities run the wrong way at every n. Note that the obstruction is *not* the germ/basin issue of remark 2.10 — the pullback there moves the evaluation point arbitrarily deep into the germ at the cost of a rational factor — it is the degree structure of the equation itself.

The consequence deserves emphasis:

transcendence — indeed, even irrationality — of $A(p_0)$ is at present not proved for a single rational $p_0 \in (0, \frac{1}{2})$.

The parameter gap

Theorem 2.16 says nothing about the map $p \mapsto A(p)$: readings (E) and (D) are open. The natural machinery is the parametrised differential Galois theory of difference equations of Hardouin and Singer [16], designed precisely to detect differential relations in auxiliary directions for linear difference equations such as $\sigma\psi = r\psi$; it does not apply as it stands, because the derivation ∂_r fails to commute with the substitution operator $\sigma\colon w \mapsto f_r(w)$, which itself moves with r. Nor does any feature of the near-critical asymptotics obstruct (E) or (D): the logarithmic corrections to (2.21) are themselves elementary functions of ε.

2.11 Inverse-symbolic evidence at eleven rational parameters

Since the theorems leave (V) open, the value question was attacked directly, by integer-relation detection. The values

$$A(p), \qquad p \in \left\{\tfrac{1}{10}, \tfrac{1}{8}, \tfrac{1}{6}, \tfrac{1}{5}, \tfrac{1}{4}, \tfrac{3}{10}, \tfrac{1}{3}, \tfrac{3}{8}, \tfrac{2}{5}, \tfrac{5}{12}, \tfrac{9}{20}\right\}, \tag{2.43}$$

were computed to more than 1150 significant digits by two mechanically independent routes — the exact product identity (2.13) iterated to convergence, and a pullback of the orbit into the Koenigs germ followed by evaluation of the Koenigs series — which agreed to at least 1192 digits at every parameter. Each value was then subjected to a PSLQ battery at ≥ 200-digit effective rigour:

- *algebraicity*: no integer polynomial relation of degree ≤ 10 and coefficient height $\leq 10^6$;
- *linearity*: no rational linear relation, at height $\leq 10^8$, against the constants π, π^2, $\ln 2$, $\ln 3$, $\ln 5$, γ, $\zeta(3)$, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, e, e^π, $\Gamma(\frac{1}{3})$, $\Gamma(\frac{1}{4})$ and Catalan's constant — also tested for $1/A$, the quasi-stationary mean itself, and for A/ε;
- *bilinearity*: no representation $A = L_1/L_2$ with L_1, L_2 integer linear forms of height $\leq 10^8$ in a twelve-constant basis, run at 460 digits;
- *multiplicativity*: no relation $A = \mathrm{e}^{q_0} 2^{q_1} 3^{q_2} 5^{q_3} 7^{q_4} \pi^{q_5}$ with rational exponents of height $\leq 10^{10}$;
- *cross-parameter*: no rational linear or multiplicative relation linking the eleven values to one another, run at 500 digits.

In all, 123 tests returned no relation — indeed no discovery-stage candidate: every PSLQ run terminated on its internal norm bound, so each null carries a genuine "no relation up to the stated height" guarantee. Throughout, the precision head-room, digits divided by basis size, exceeded every height cap by at least eight orders of magnitude, so a returned relation would have been meaningful rather than numerical noise.[1]

This is finite evidence: it excludes low-complexity closed forms over the tested constants, and nothing more. A closed form over untested constants — $\Gamma(\frac{1}{7})$, say, or periods of higher-genus curves — or one of height above the caps would have escaped the search.

2.12 Computing $A(p)$ in practice

In the absence of an exact formula we are left with approximation, and the two regimes identified above suggest how to split the work.

Over most of the interval, direct iteration is entirely satisfactory: the infinite product (2.13) converges quickly, its partial products bound $A(p)$ from above at every order, the tail bound (2.15) supplies a matching lower bound and hence a stopping certificate, and modest numbers of iterations

[1] Computations in `mpmath` at working precisions of 310–1200 digits; the values, scripts and full per-test log are archived with the project sources and are not reproduced in this thesis.

give many digits. The difficulty is confined to a neighbourhood of $p = \frac{1}{2}$, and the binding constraint is that as the process is defined ever closer to criticality its lifetime, though still finite, grows enormously, and S_n recedes to zero ever more slowly. The iteration must be carried to a generation at which S_n is essentially unchanging, and table 2.1 shows what that costs: about 10^2 iterations at $p = 0.2$, 10^3 at $p = 0.45$, but nearly 5×10^5 at $p = 0.4999$.

The remedy is to hand the near-critical region to the asymptotic (2.21), which is exact in the limit and cheap everywhere:

$$A(p) \approx \begin{cases} \dfrac{1}{2} \displaystyle\prod_{k=1}^{N} \left(1 - \frac{S_k}{2}\right), & p < p_*, \quad N \gg 1, \\ 2 - 4p, & p \geq p_*, \end{cases} \tag{2.44}$$

with the crossover p_* chosen just below $\frac{1}{2}$. Remark 2.7 indicates where to put it. The relative error of the asymptotic is about $\varepsilon \log(1/\varepsilon)$ by theorem 2.5, so at $p_* = 0.4999$ (that is $\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate to about two parts in 10^3, while the product still converges in a few hundred thousand iterations. Any p_* in the range 0.499 to 0.4999 gives a scheme accurate to better than one per cent throughout, and the two branches can be blended over a narrow window if continuity of the derivative matters.

First, use the product rather than the raw ratio $S_n/(2p)^n$: only the product comes with the certificate (2.15).

Second, stop on the certificate, not on apparent convergence. In the near-critical regime successive floating-point iterates cease to differ long before the tail is genuinely negligible, and a stopping rule based on observed change silently returns the wrong answer. The bracket in (2.15) costs one extra multiplication and removes the failure mode.

Third, if the quantity actually needed is the quasi-stationary mean $1/A$ rather than A itself — as it usually is — then the series (2.17) computes it directly, and its partial sums increase to the answer, giving the complementary one-sided bound.

This is the recipe used for the numerical values quoted throughout this chapter and in the mathematical introduction, including table 2.1.

2.13 Conclusion

The constant $A(p)$ organises the quasi-stationary behaviour of subcritical binary Galton–Watson branching: the limiting conditional mean is $1/A(p)$, the limiting conditional variance is (2.7), and both are functions of p and $A(p)$ alone. This chapter has shown that the constant is well defined (proposition 2.1), exactly representable as an infinite product (2.13) and as a series (2.17), confined by proposition 2.3 between $\frac{1}{2}A_{\mathrm{c}}(p)$ and $2\varepsilon/(1+2\varepsilon)$ and by proposition 2.4 below $\frac{1}{2}$, asymptotically equal to 2ε near criticality with a relative correction of order $\varepsilon \log(1/\varepsilon)$ (theorem 2.5), and computable to arbitrary accuracy with a certificate by the hybrid scheme (2.44). For every purpose this thesis has, $A(p)$ is a usable object.

What it is not is elementary. By proposition 2.9 the constant is twice the basin value at $\frac{1}{2}$ of the Koenigs linearising function of the logistic map at multiplier $r = 2p$, and by theorem 2.16 that function is hypertranscendental in its own variable for every r in the subcritical window — with no exceptional parameters to exclude, since differentially algebraic Schröder functions exist only over repelling fixed points, and the window is entirely attracting.

What the theorem does not decide is set out in section 2.10.

Two problems are therefore left open by this chapter.

1. *The value problem.* Is $A(p_0)$ irrational for some, or every, rational $p_0 \in (0, \frac{1}{2})$? A positive answer for a single parameter would be the first arithmetic result about this constant.
2. *The parameter problem.* Is $p \mapsto A(p)$ differentially algebraic in p? This is the reading under which a "formula for $A(p)$" would most naturally be sought, and it is precisely the reading theorem 2.16 does not address. A negative answer would close the matter; a positive one would be more surprising still.

The exceptional elementary linearisations at $r = 2$ and $r = 4$, both over repelling fixed points and both outside the branching window, are derived in section 2.A and located inside the Becker–Bergweiler classification there.

2.A Elementary Koenigs functions at $r = 2$ and $r = 4$

These cases lie outside subcritical branching, where $r = 2p < 1$, but they complete the classical picture of integrable logistic linearisations and they show what a solvable case looks like. At both parameters the fixed point at the origin is *repelling*, so the functions below are Poincaré-type linearisations rather than attracting Koenigs maps; remark 2.17 explains why that is the decisive point rather than an incidental one.

2.A.1 The monomial case, $r = 2$

For $r = 2$ we have $f_2(x) = 2x(1 - x)$, and the fixed point 0 has multiplier $f_2'(0) = 2$. The affine change $u = 1 - 2x$ conjugates the dynamics to the monomial map: since

$$1 - 2f_2(x) = 1 - 4x + 4x^2 = (1 - 2x)^2,$$

the substitution gives $u_{n+1} = u_n^2$, which the logarithm linearises because $\ln(u^2) = 2 \ln u$. Returning to x and imposing the normalisation of definition 2.8,

$$\psi_2(x) = -\tfrac{1}{2} \ln(1 - 2x), \tag{2.45}$$

which is readily verified:

$$\psi_2\big(f_2(x)\big) = -\tfrac{1}{2} \ln\big((1 - 2x)^2\big) = -\ln(1 - 2x) = 2\psi_2(x),$$

with $\psi_2(0) = 0$ and $\psi_2'(0) = 1$.

2.A.2 The Chebyshev case, $r = 4$

For $r = 4$ we have $f_4(x) = 4x(1 - x)$, with multiplier $f_4'(0) = 4$. Putting $x = \sin^2 \theta$,

$$f_4(x) = 4 \sin^2 \theta \cos^2 \theta = (2 \sin \theta \cos \theta)^2 = \sin^2(2\theta),$$

so the map is a doubling $\theta \mapsto 2\theta$ in the θ coordinate. The multiplier at the origin is 4 rather than 2, which suggests that the linearisation involves θ^2; and indeed

$$\psi_4(x) = \big(\arcsin \sqrt{x}\big)^2. \tag{2.46}$$

Using $\arcsin\big(2u\sqrt{1 - u^2}\big) = 2 \arcsin u$ with $u = \sqrt{x}$, on an appropriate real interval, gives $\psi_4(f_4(x)) = (2 \arcsin \sqrt{x})^2 = 4\psi_4(x)$; and as $x \to 0$, $\arcsin \sqrt{x} \approx \sqrt{x}$, so $\psi_4(x) \approx x$ and $\psi_4'(0) = 1$.

2.A.3 Affine conjugacy to $z^2 + c$

Lemma 2.18. *The logistic map $f_r(x) = rx(1-x)$ is affinely conjugate to $P_c(z) = z^2 + c$ with*

$$c = c(r) := \frac{2r - r^2}{4}. \tag{2.47}$$

Proof sketch. Seek $h(x) = ax+b$ with $h \circ f_r \circ h^{-1} = P_c$. The substitution $z = r(\frac{1}{2} - x)$ — equivalently $x = -z/r + \frac{1}{2}$, for $r \neq 0$ — yields a monic quadratic; if $x' = f_r(x)$ and $z' = r(\frac{1}{2} - x')$, direct substitution gives $z' = z^2 + c$ with c as in (2.47). □

Under this conjugacy, $r = 2$ corresponds to $c = 0$ (the monomial z^2) and $r = 4$ to $c = -2$ (Chebyshev type; the value $c = -2$ is also reached at $r = -2$). Neither exceptional value lies in the image of $r \in (0, 1)$, which is $c \in (0, \frac{1}{4})$.

2.A.4 The worked examples inside the Becker–Bergweiler classification

Theorem 2.14 says that a differentially algebraic Schröder function exists only over a repelling fixed point, and is then a Möbius transformation of $\exp(\alpha t^{\varrho})$, of $\cos(\alpha t^{\varrho} + \beta)$, or of Weierstrass-$\wp$ material, with the underlying map Möbius-conjugate to $z^{\pm d}$, to $\pm T_d$, or to a Lattès map. The two derivations above realise the first two families exactly.

Case $r = 2$. Inverting (2.45), the relation $t = -\frac{1}{2}\ln(1 - 2x)$ gives

$$\varphi_2(t) = \psi_2^{-1}(t) = \frac{1 - \mathrm{e}^{-2t}}{2},$$

an affine — hence Möbius — image of $\exp(-2t)$: the family $\exp(\alpha t^{\varrho})$ with $\varrho = 1$, $\alpha = -2$. Correspondingly the substitution $u = 1 - 2x$ conjugates f_2 to the monomial $u \mapsto u^2$, and the linearised fixed point has multiplier 2, which is repelling, as theorem 2.14(i) requires.

Case $r = 4$. Inverting (2.46), the relation $t = (\arcsin\sqrt{x})^2$ gives

$$\varphi_4(t) = \psi_4^{-1}(t) = \sin^2\sqrt{t} = \frac{1 - \cos(2\sqrt{t})}{2},$$

a Möbius image of $\cos(\alpha t^{\varrho} + \beta)$ with $\varrho = \frac{1}{2}$, $\alpha = 2$, $\beta = 0$; the fractional exponent is admissible precisely because $\cos(2\sqrt{t})$ is an entire function of t. Correspondingly $u = 1 - 2x$ conjugates f_4 to $T_2(u) = 2u^2 - 1$, the second Chebyshev polynomial, and the multiplier at the linearised fixed point is 4, again repelling.

Both examples therefore sit exactly where theorem 2.14 says all differentially algebraic Schröder functions must sit: over repelling fixed points, in the exponential and cosine families.

2.B Catalogue of exploratory formulae

This appendix records the remaining output of the closed-form searches described in section 2.7. The expressions below are approximations only.

The constrained rational candidate

Polynomial and rational searches can be constrained to the known endpoints $A(0) = \frac{1}{2}$ and $A(\frac{1}{2}) = 0$. One of the more economical rational candidates was $\hat{A}_1$ of (2.31), reproduced here for comparison:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}, \qquad \hat{A}_1'(\tfrac{1}{2}) = -\frac{2}{0.55} \approx -3.636.$$

It passes through both endpoints and misses the critical slope -4 required by remark 2.6.

Unconstrained symbolic tree search

A less restricted symbolic tree search produced many longer formulae of similar visual quality. Two representative outputs were

$$\hat{A}_2(p) = 0.31696448624690005 \log\Bigg(\Bigg|\cos\Bigg(\bigg(\frac{1}{3.19801133824532}\bigg)^{1/4} + p\Big[\sin\Big(\cos\big(|p|(0.621865696665606 - p)\cos(\mathrm{e}^{\mathrm{e}^{p}})\big)\Big) + |p|\Big]\Bigg)\Bigg|\Bigg) + 0.5982282661889831 \quad (2.48)$$

and $\hat{A}_3$ of (2.32). Both reproduce the target curve to within the thickness of a plotted line across the whole interval (figure 2.3), and neither is interpretable in any useful sense.

Spliced surrogates

Forcing the correct critical branch by splicing gives $\hat{A}_4$ of (2.33), which is a workable practical surrogate and nothing more.

2.C Figure candidate gallery (temporary)

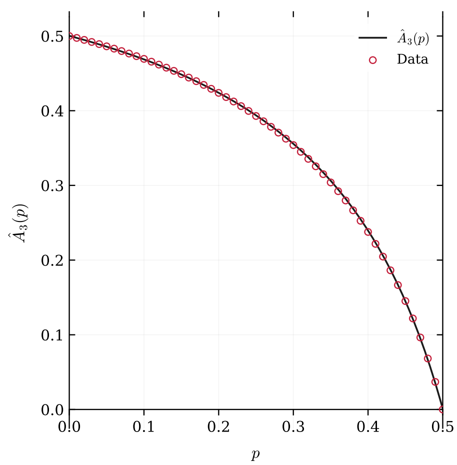
This appendix is a temporary asset-selection aid and forms no part of the chapter's argument. It displays every figure binary collected from the Claude, Qwen, Grok and Codex merges, together with the assets currently used in this project, which carry the `best` tag. Each panel is labelled with its candidate filename: the prefix is the source, and the remainder is the path the file occupied in that source's `figures/` tree, with directory separators written as double underscores. Variants sharing a filename stem are grouped so that they can be compared side by side. Nothing has been dropped as redundant, so byte-identical binaries appear more than once by design. After selection, this appendix and the unused candidates will be removed.

There are 68 candidates here in 19 groups: 11 with more than one variant, and 8 one-offs.

Index of groups

Comparison groups: two or more variants

Group: `a3_hat_plot`



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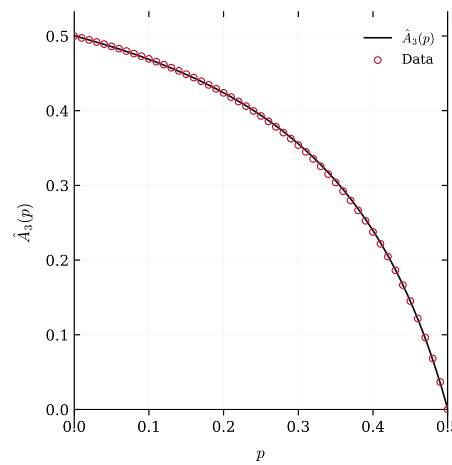

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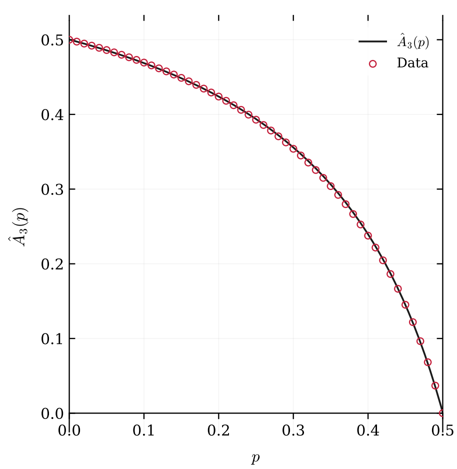

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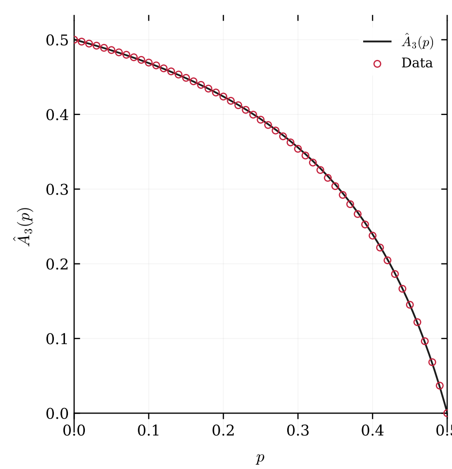

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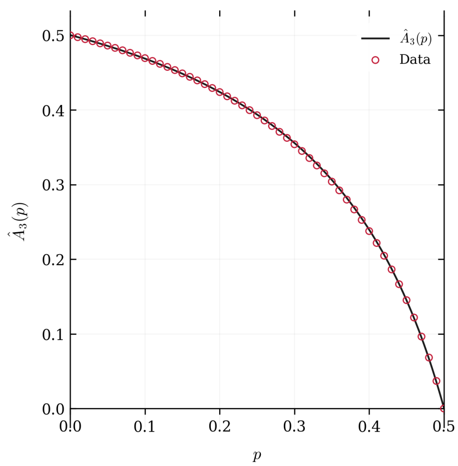
Figure 2.8: Candidates for group `a3_hat_plot` (9 variants), part 1 of 3. Filenames are printed beneath the panels.



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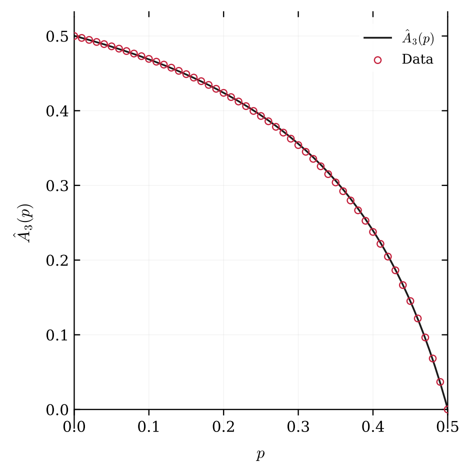

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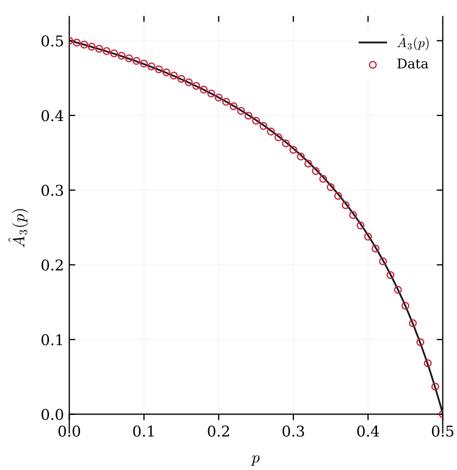

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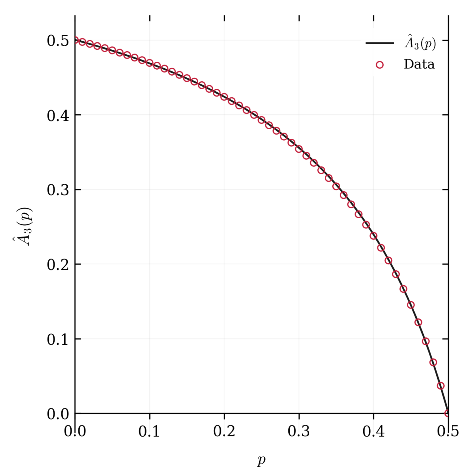

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Figure 2.9: Candidates for group `a3_hat_plot` (9 variants), part 2 of 3. Filenames are printed beneath the panels.

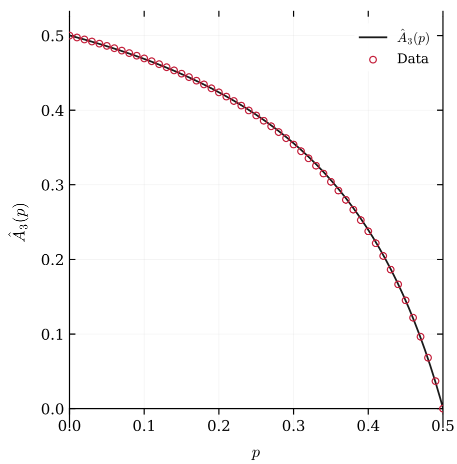

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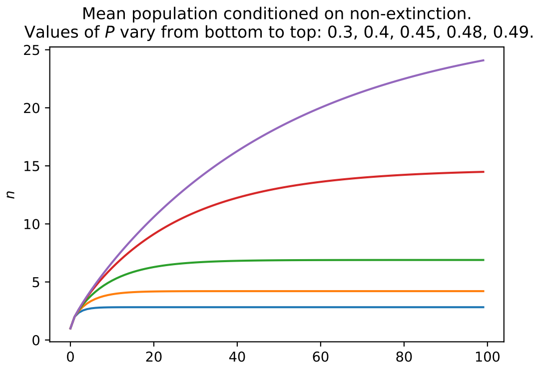
Figure 2.10: Candidates for group `a3_hat_plot` (9 variants), part 3 of 3. Filenames are printed beneath the panels.

Group: `conditionalmean`



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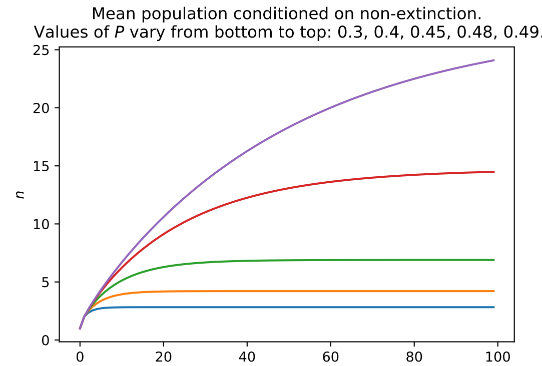

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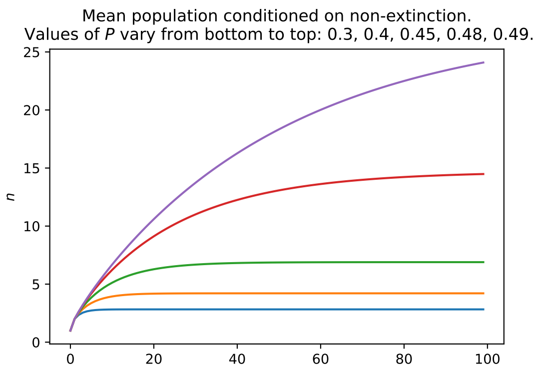

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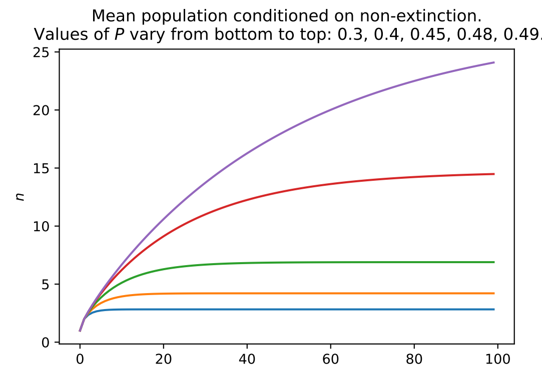

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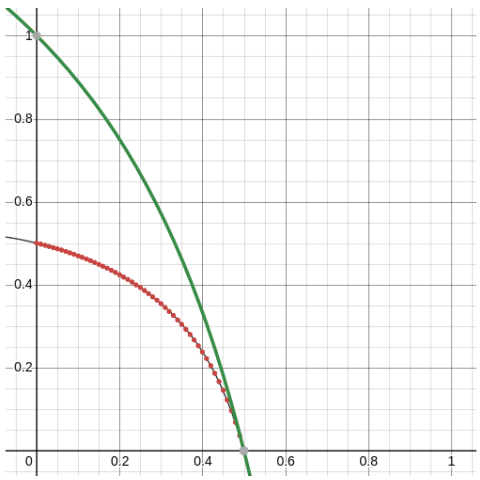
Figure 2.11: Candidates for group `conditionalmean` (4 variants). Filenames are printed beneath the panels.

Group: `dtcta`



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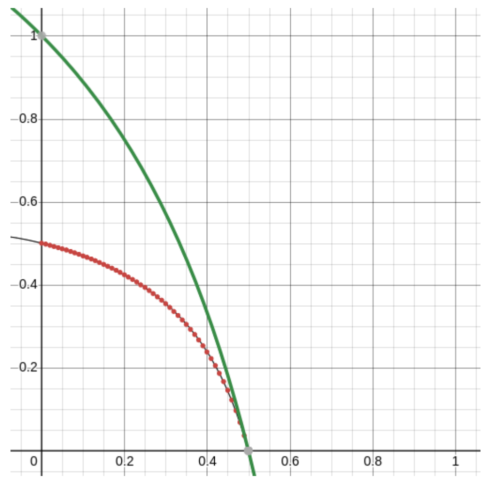

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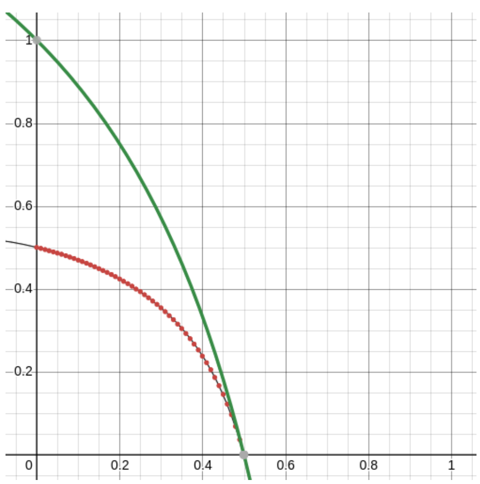

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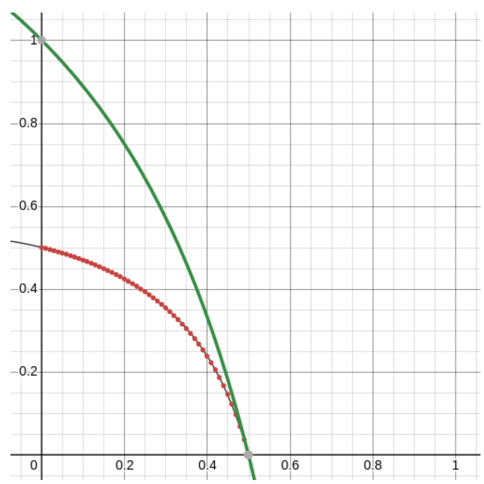

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Figure 2.12: Candidates for group `dtcta` (5 variants), part 1 of 2. Filenames are printed beneath the panels.

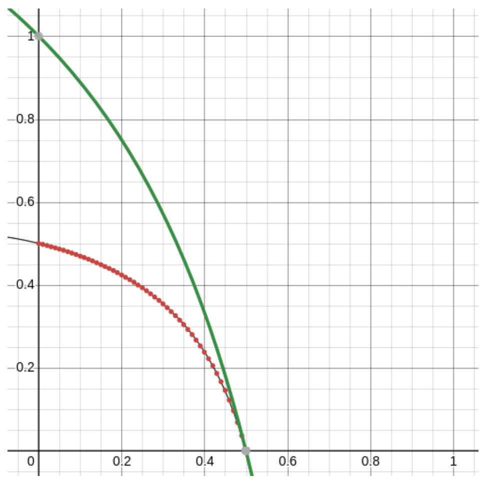

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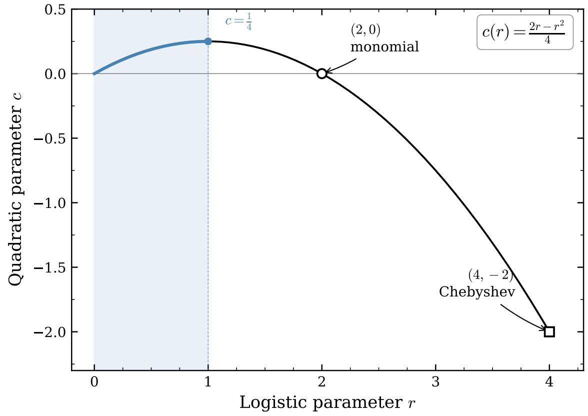
Figure 2.13: Candidates for group dtcta (5 variants), part 2 of 2. Filenames are printed beneath the panels.

Group: `figure1_parameter_conjugacy`



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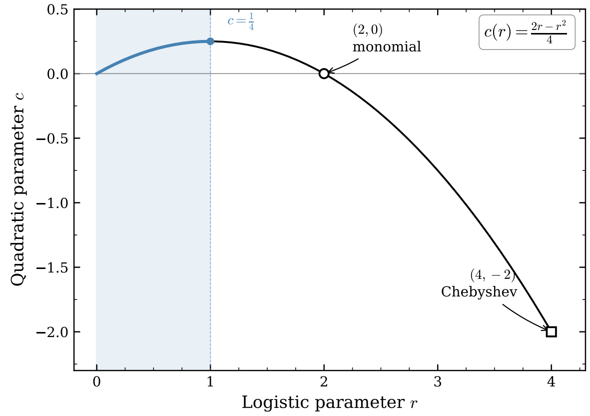

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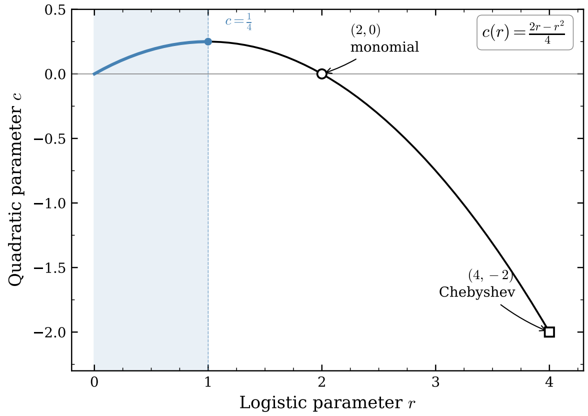

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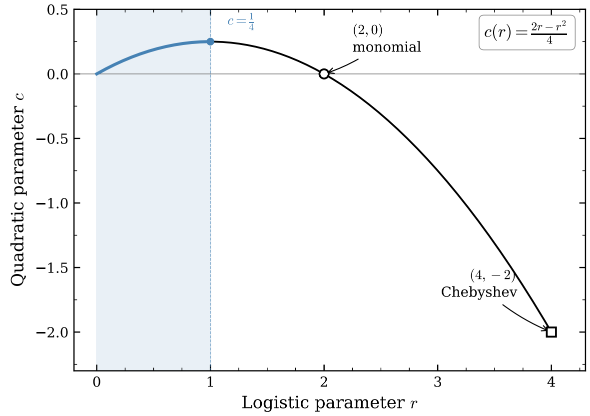

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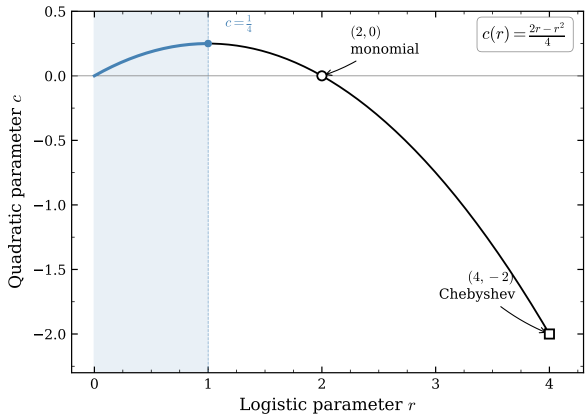
Figure 2.14: Candidates for group `figure1_parameter_conjugacy` (7 variants), part 1 of 2. Filenames are printed beneath the panels.



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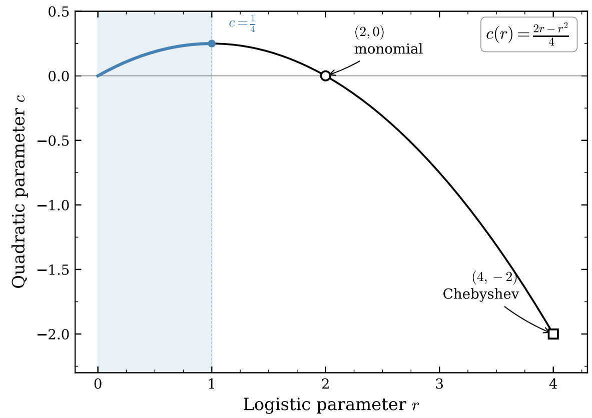

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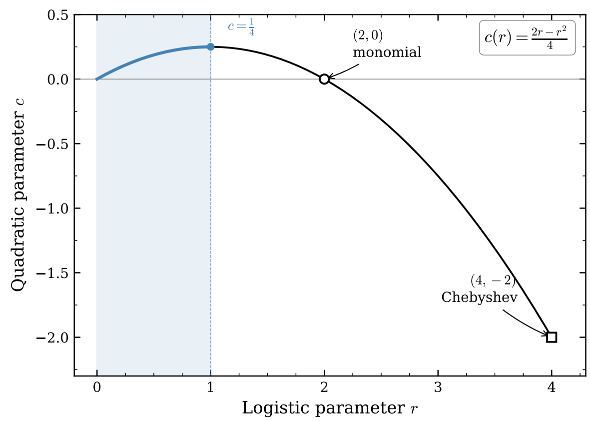

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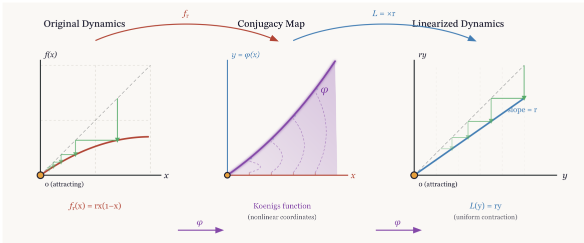
Figure 2.15: Candidates for group `figure1_parameter_conjugacy` (7 variants), part 2 of 2. Filenames are printed beneath the panels.

Group: `figure2_koenigs_linearization`



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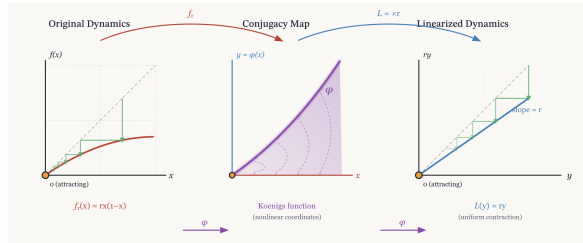

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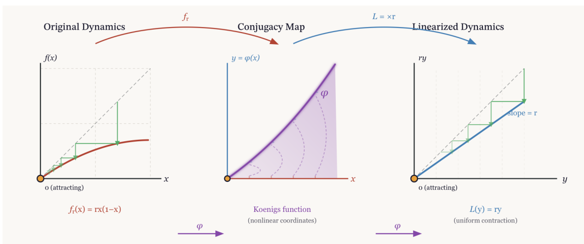

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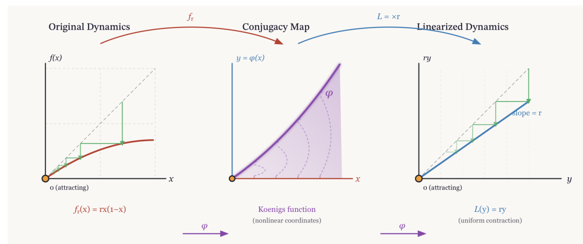

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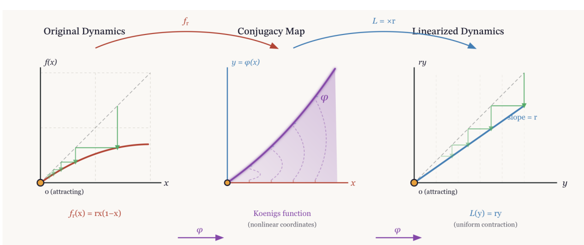
Figure 2.16: Candidates for group `figure2_koenigs_linearization` (7 variants), part 1 of 2. Filenames are printed beneath the panels.



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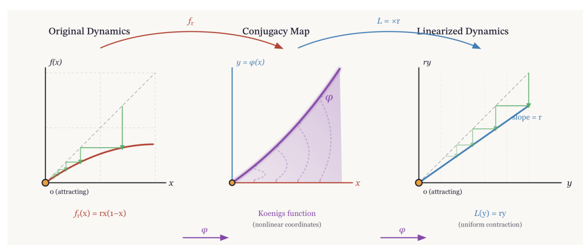

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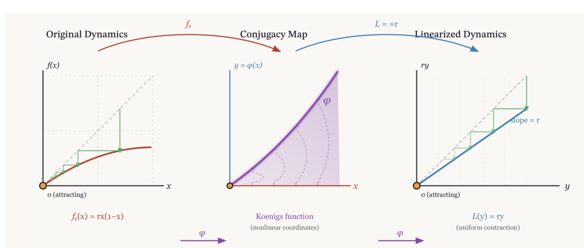

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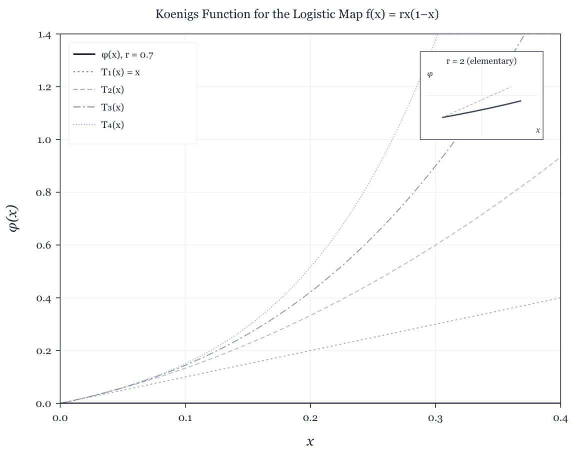
Figure 2.17: Candidates for group `figure2_koenigs_linearization` (7 variants), part 2 of 2. Filenames are printed beneath the panels.

Group: `figure3_numerical_koenigs`



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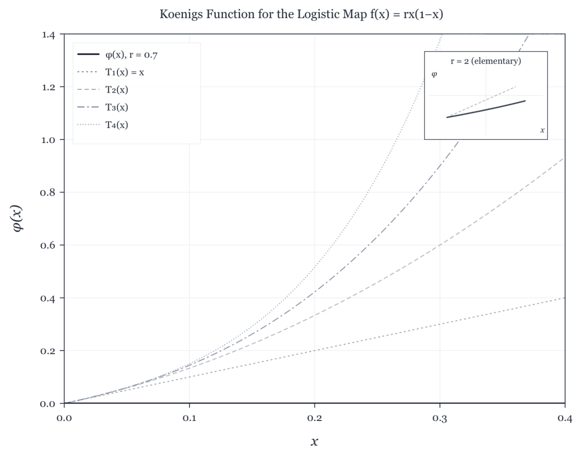

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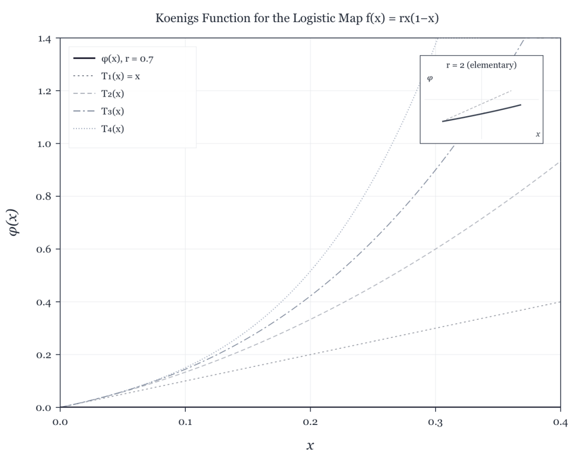

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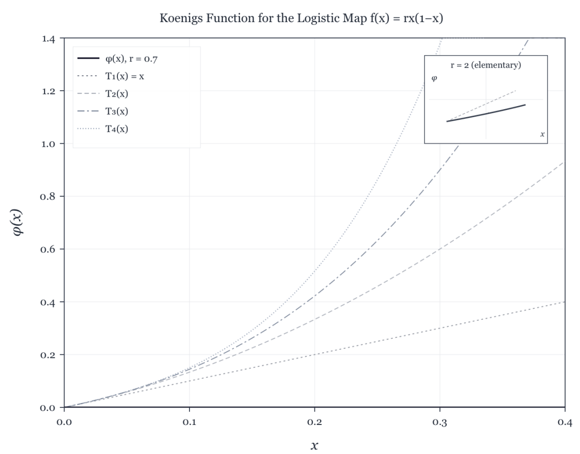

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Figure 2.18: Candidates for group `figure3_numerical_koenigs` (5 variants), part 1 of 2. Filenames are printed beneath the panels.

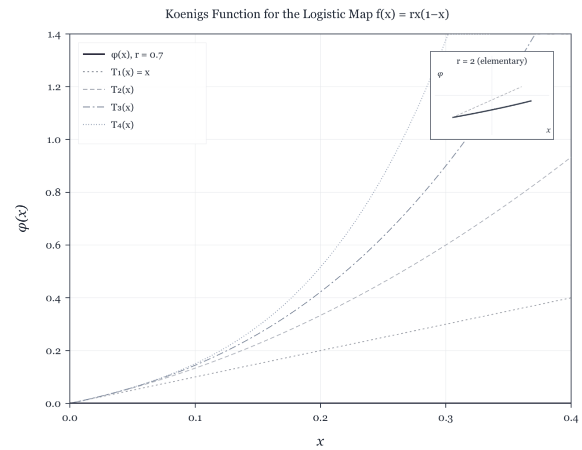

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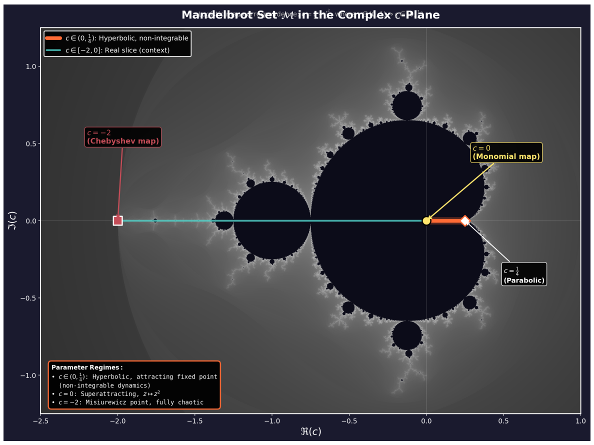
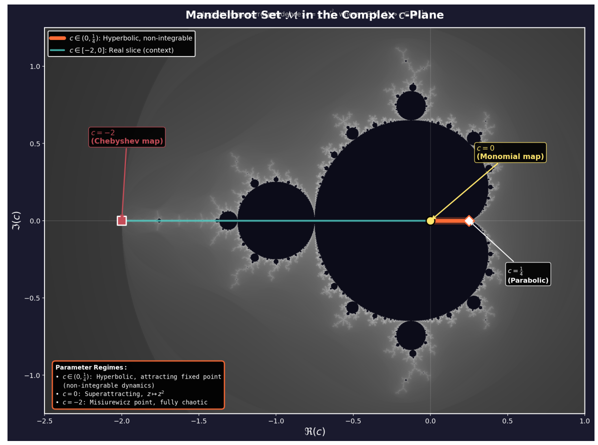
Figure 2.19: Candidates for group `figure3_numerical_koenigs` (5 variants), part 2 of 2. Filenames are printed beneath the panels.

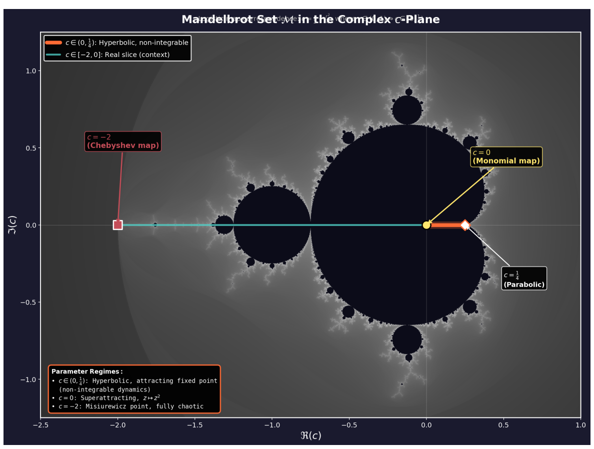
Group: `figure4_mandelbrot_context`



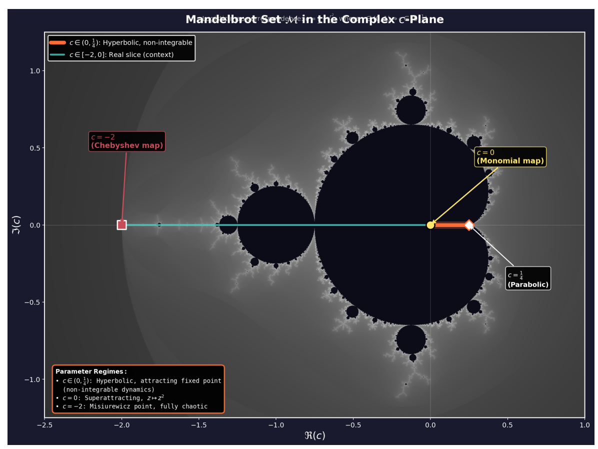
`best__figure4_mandelbrot_context.png`



`claude__figure4_mandelbrot_context.png`

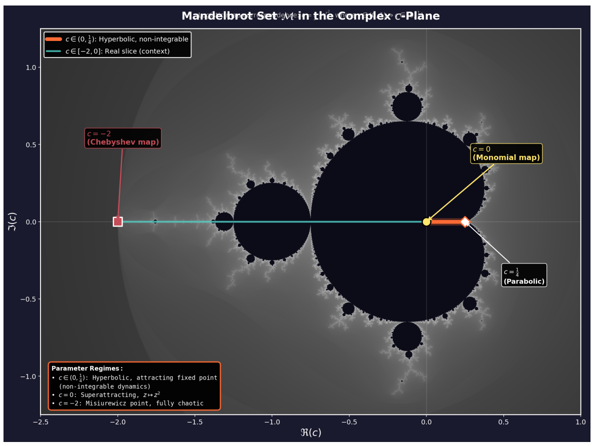


`qwen__figure4_mandelbrot_context.png`

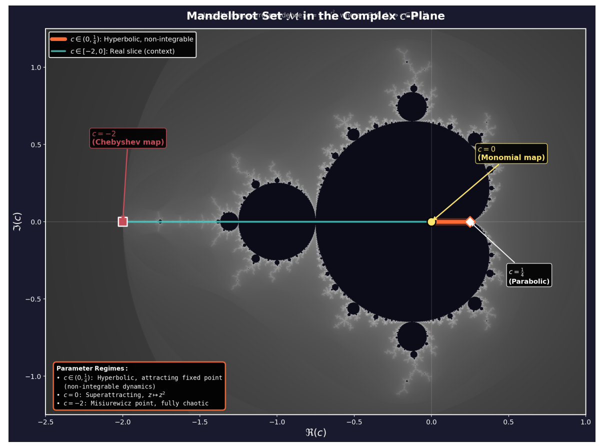


`grok__IMG_ch3__figure4_mandelbrot_context.png`

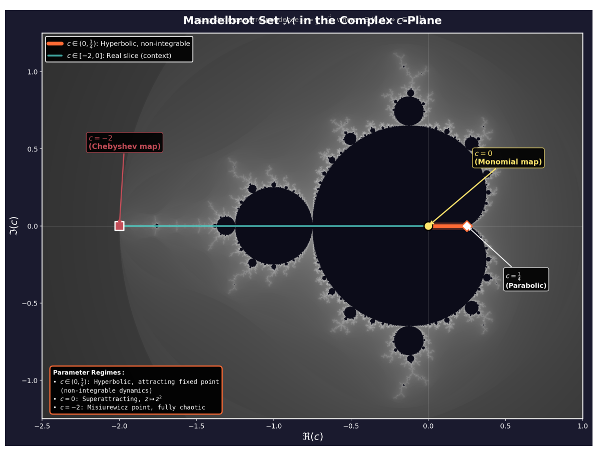
Figure 2.20: Candidates for group `figure4_mandelbrot_context` (7 variants), part 1 of 2. Filenames are printed beneath the panels.



grok__figure4_mandelbrot_context.png



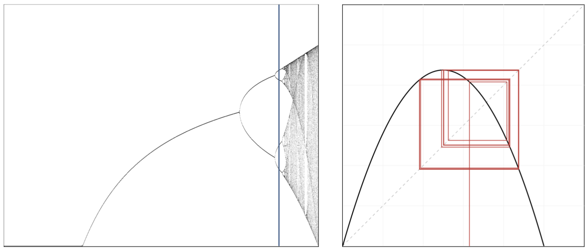
codex__IMG_ch3__figure4_mandelbrot_context.png



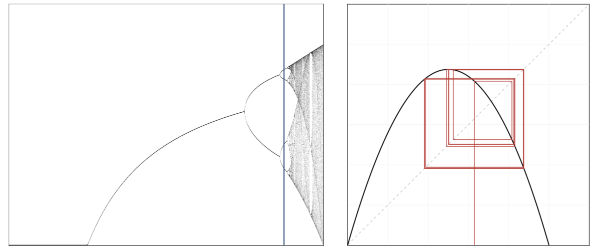
codex__figure4_mandelbrot_context.png

Figure 2.21: Candidates for group `figure4_mandelbrot_context` (7 variants), part 2 of 2. Filenames are printed beneath the panels.

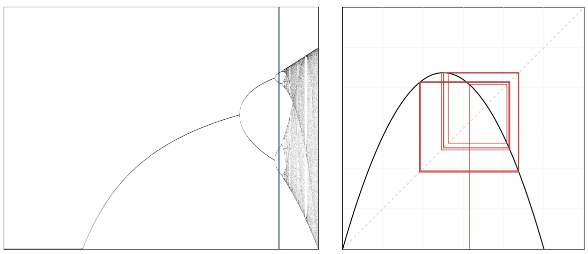
Group: `period_double`



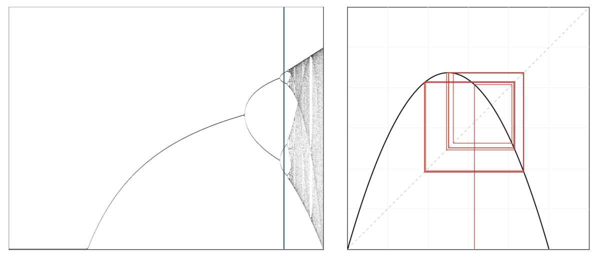
`best__period_double.png`



`claude__period_double.png`

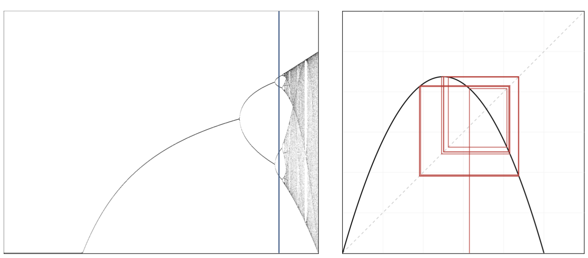


`qwen__period_double.png`

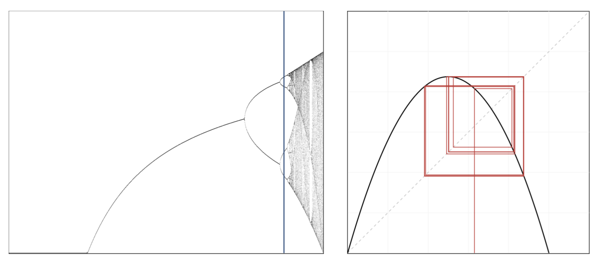


`grok__IMG_ch3__period_double.png`

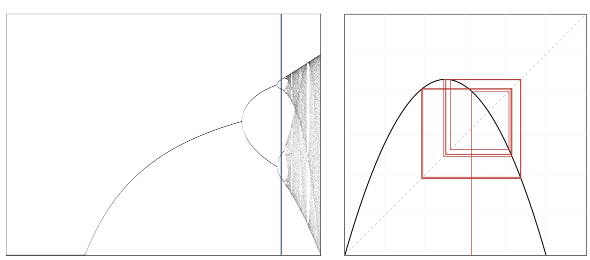
Figure 2.22: Candidates for group `period_double` (7 variants), part 1 of 2. Filenames are printed beneath the panels.



`grok__period_double.png`



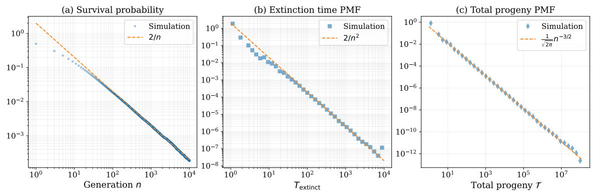
`codex__IMG_ch3__period_double.png`



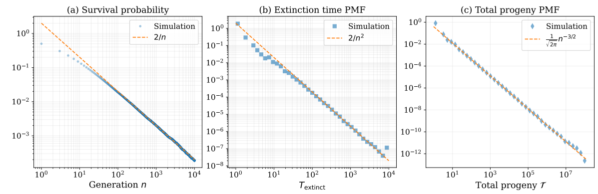
`codex__period_double.png`

Figure 2.23: Candidates for group `period_double` (7 variants), part 2 of 2. Filenames are printed beneath the panels.

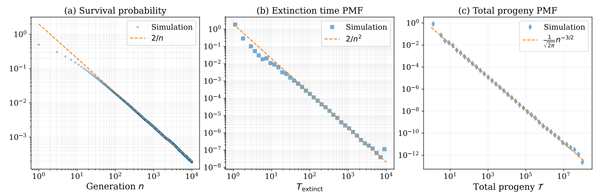
Group: `power_law_fixed`



`grok__power_law_fixed.pdf`



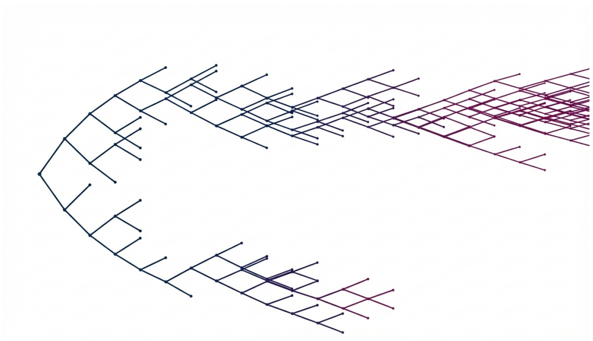
`codex__IMG_ch3__power_law_fixed.pdf`



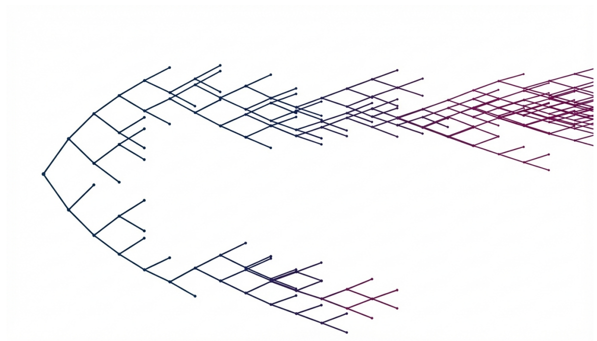
`codex__power_law_fixed.pdf`

Figure 2.24: Candidates for group `power_law_fixed` (3 variants). Filenames are printed beneath the panels.

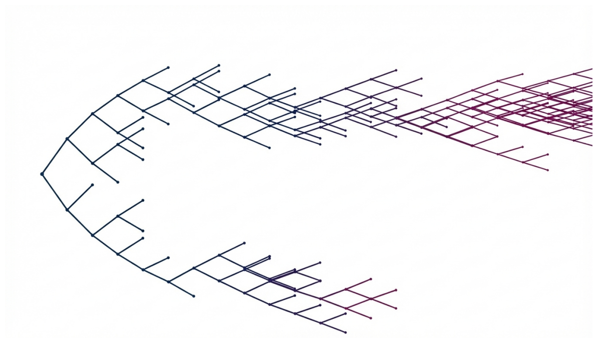
Group: `simplegwvis`



grok__simpleGWvis.png



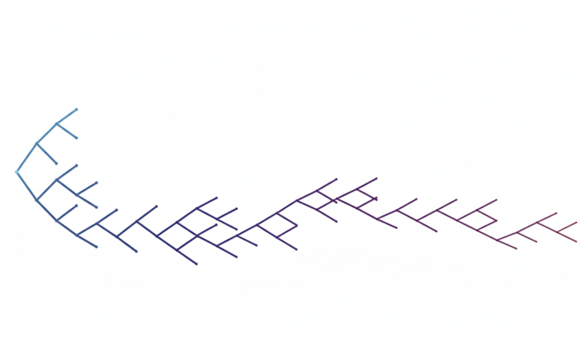
codex__IMG_ch3__simpleGWvis.png



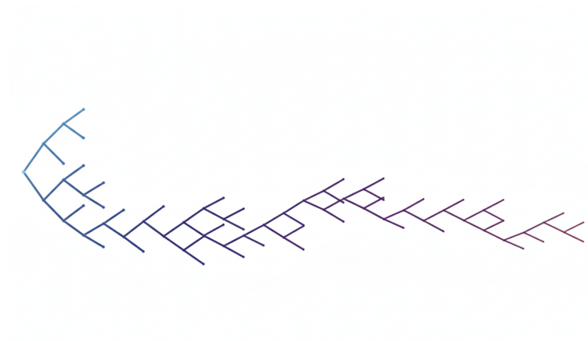
codex__simpleGWvis.png

Figure 2.25: Candidates for group `simplegwvis` (3 variants). Filenames are printed beneath the panels.

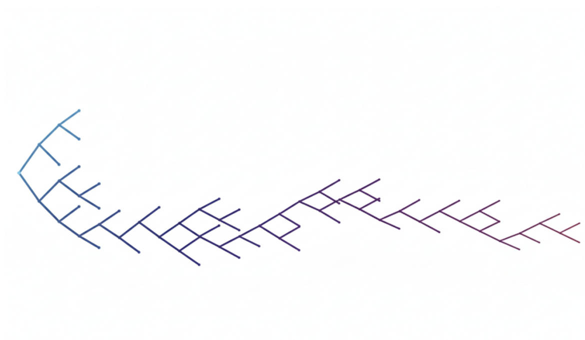
Group: `subgwvis`



grok__subGWvis.png



codex__IMG_ch3__subGWvis.png

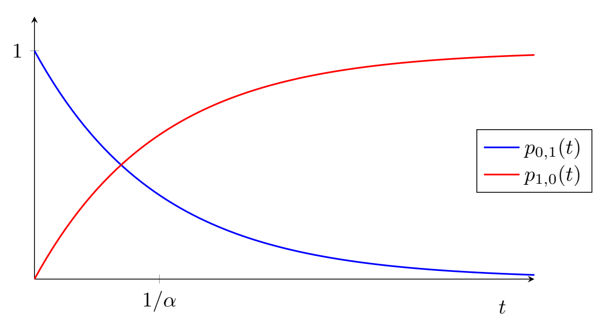


codex__subGWvis.png

Figure 2.26: Candidates for group subgwvis (3 variants). Filenames are printed beneath the panels.

Unique assets: a single source

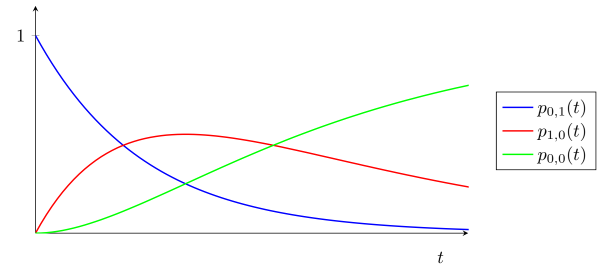
Group: abs1



codex__IMG_ch3__abs1.pdf

Figure 2.27: Candidates for group abs1 (1 variant). Filenames are printed beneath the panels.

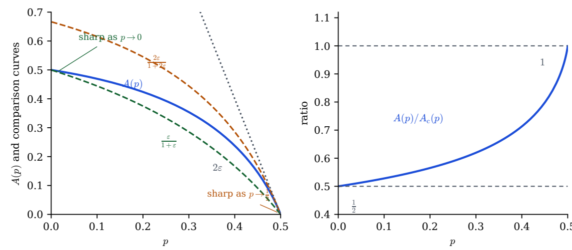
Group: abs2



codex__IMG_ch3__abs2.pdf

Figure 2.28: Candidates for group abs2 (1 variant). Filenames are printed beneath the panels.

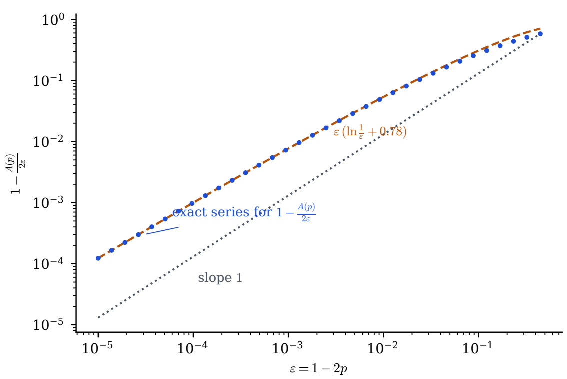
Group: ap_bounds_ratio



qwen__Ap_bounds_ratio.pdf

Figure 2.29: Candidates for group ap_bounds_ratio (1 variant). Filenames are printed beneath the panels.

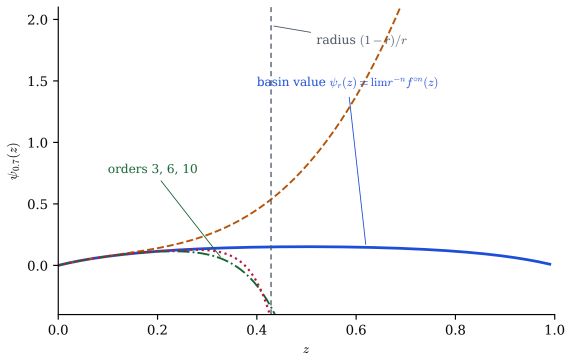
Group: ap_nearcrit



qwen__Ap_nearcrit.pdf

Figure 2.30: Candidates for group ap_nearcrit (1 variant). Filenames are printed beneath the panels.

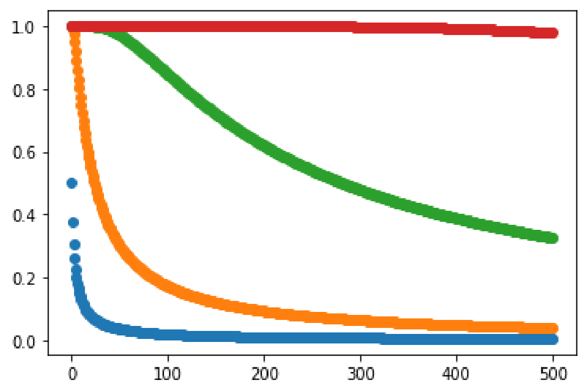
Group: koenigs_domain



`qwen__koenigs_domain.pdf`

Figure 2.31: Candidates for group `koenigs_domain` (1 variant). Filenames are printed beneath the panels.

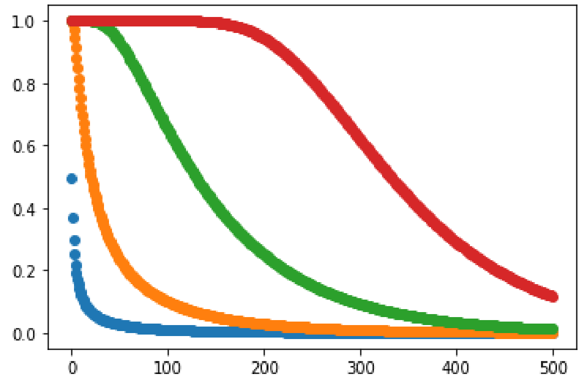
Group: `kvals`



`codex__IMG_ch3__kvals.png`

Figure 2.32: Candidates for group `kvals` (1 variant). Filenames are printed beneath the panels.

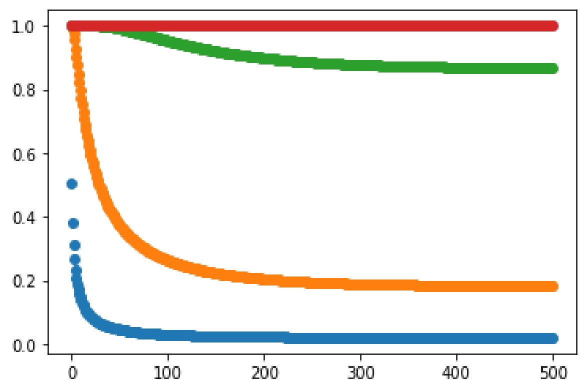
Group: `kvals495`



`codex__IMG_ch3__kvals495.png`

Figure 2.33: Candidates for group `kvals495` (1 variant). Filenames are printed beneath the panels.

Group: `kvals505`



`codex__IMG_ch3__kvals505.png`

Figure 2.34: Candidates for group `kvals505` (1 variant). Filenames are printed beneath the panels.

Candidates that failed the integrity check

None. Every staged candidate passed the header and `pdfinfo` checks, and every one of them is displayed above.

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